

A Quantum Structural Triad: Fluctuations, Entropy, and Correlations as Interdependent Primitives

Ovidiu Tataru

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Abstract

We propose a structural condition in quantum theory: the product of local fluctuation (via $\sqrt{F_Q}$), von Neumann entropy, and quantum mutual information *for a subsystem A* forms a *quantum structural triad* that, under locality, genuine cross-cut coupling, and global non-stationarity (H1–H3), does not vanish in time-average (within H1–H3 and positive-mean conditions). The triad serves as a minimal indicator of structural integrity for a quantum state to persist, evolve, and mediate phenomena. From these hypotheses we derive two quantitative thresholds—a Hamiltonian-capped form controlled by minimal local energy scales and subsystem dimensions, and a speed-based form controlled by Bures/QFI speed floors and the density of an active epoch. The formulation is intensive and systemic: a normalized, dimensionless triad enables scale-free comparison, and cut-set averages diagnose the integrity of the global state rather than the autonomy of its parts. We illustrate the framework on pure and mixed states, give applications to qubit screening, memory stability, and error-correction prioritization, and outline extensions to open systems via entropy production, as well as a possible route to algebraic QFT. Extensive numerical studies (TEBD dynamics, integrable/nonintegrable baselines, eigenstate checks, and a topological prototype) support the robustness of the thresholds. Overall, the results provide falsifiable, hypothesis-qualified criteria for assessing structural viability in quantum technologies.

Status. This manuscript presents a hypothesis-qualified framework. Results are necessary (not sufficient) within H1–H3 and full-rank epochs, and do not claim a fundamental derivation beyond the stated assumptions. We establish necessary lower bounds via a probabilistic route (Paley-Zygmund with ergodic averaging), using the QFI-variance relation only to translate units. Numerical validations (Annex A15) confirm empirical exceedance rates above the PZ floor across synthetic and physics-proxy time series. Results are necessary (not sufficient) under H1–H3; topological/AQFT remarks are conjectural and confined to the annex.

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1 Introduction

Quantum theory imposes fundamental constraints on what qualifies as a physically viable subsystem—understood here as a resolvable region within an interacting system. Unless otherwise noted, we interpret local generators as time-scaled (i.e., $H/\hbar$ acts as the time generator), so that $\sqrt{F_Q}$ carries units of inverse time. This convention keeps all thresholds and floors dimensionally consistent in SI, and matches the response usage in the companion derivation. From the uncertainty principle to modern information-theoretic bounds[1, 2], it has long been clear that fluctuations, entropy, and correlations cannot all vanish simultaneously without undermining the internal consistency of the theory. Each captures a distinct aspect of quantum structure: energy fluctuations quantify sensitivity to local generators[3, 4], von Neumann entropy measures local information content[5, 6], and mutual information encodes connectivity across subsystems[7, 8]. While each has been widely studied in isolation or in pairs—as resources for metrology[9, 10], thermodynamics[11, 12], or information processing[13]—their collective interdependence has remained uncharacterized.

Existing work has established important pairwise constraints: generalized speed limits connect quantum Fisher information to entropy production[14, 16], fluctuation–speed relations bound entropy growth in open systems[17, 18], and entanglement persists under noise through correlation–entropy trade-offs[19, 20]. Yet none of these results capture the essential *three-way structural interdependence* that we derive under the stated assumptions to be important for system viability.

Our main result is that $\Pi(t)$, while possibly vanishing at isolated instants, cannot vanish on average across subsystems and over time, *within the stated hypotheses* (H1–H3 and positive-mean conditions; conditional on $\mathbb{E}[\Pi_A] > 0$), in any locally coupled, physically viable quantum system. This suggests the triad as a *minimal structural threshold for subsystem viability* within the stated assumptions.

Pointwise vanishing may occur, but systemic persistence is expected under these hypotheses. This motivates a minimal structural threshold below which a subsystem becomes *quantum-structurally inert*[10, 21]: unlikely to sustain dynamics, store information, or mediate interactions. Conversely, persistent nonvanishing of the triad indicates that a subsystem possesses the minimal structural integrity required to remain quantum-mechanically relevant.

The novelty lies not in recognizing that these three quantities matter, but in deriving explicit, unit-correct *quantitative thresholds* that determine how much of each is required and how they trade off against one another. These thresholds scale nontrivially with system size and minimal energy scales[22, 23], providing a falsifiable criterion for subsystem viability. Unlike heuristic measures of coherence, our results are developed from standard first-principles tools, yielding both theoretical universality and practical tools for quantum technologies.

Motivation and position in the literature. Classical and quantum speed limits link fluctuation to distinguishability, thermodynamic relations link entropy production to dynamics, and information-theoretic results bound correlations across cuts. Prior work has established many pairwise constraints (fluctuation–speed, speed–entropy production, correlation–entropy), but not a unified product threshold that operates at the subsystem level under locality. Our contribution is to integrate these strands and provide hypothesis-qualified, unit-consistent thresholds for the joint fluctuation–entropy–correlation product, with intensive normalization for system-size comparisons.

Related work (brief). *What is reused vs. novel.* Classical speed limits, QFI–variance (Identity 4), entropy–production (Spohn) and correlation/area-law results are *not* rederived here; they are used as standard inputs or unit bridges. The novelty lies in the *subsystem-level product threshold* (fluctuation $\times$ entropy $\times$ correlation), its intensive normalization, and the two structural thresholds (Hamiltonian-capped and speed-based) that couple the three legs under locality and non-stationarity.

Relation to established speed and thermodynamic bounds. Our speed based inequality complements quantum speed limit results (e.g., Deffner & Lutz [14]) by coupling the QFI (speed) leg with entropic and correlational legs into a product threshold at the subsystem level. We retain the QFI variance relation as Identity 4 solely as a unit bridge; it is not used as a proving step in the Paley Zygmund route. Likewise, our statements are consistent with entanglement thermodynamics and entropy production relations (e.g., Spohn [12]); here they appear only as connections, not rederivations under new notation.

Novelty and contributions. We introduce and examine a hypothesis-qualified *product threshold* that synthesizes the fluctuation (QFI/variance), entropic, and correlational legs. We provide two complementary forms: (i) a Hamiltonian-capped bound governed by minimal operator norms and subsystem dimensions, and (ii) a speed-based bound governed by Bures/QFI speed floors and active-epoch density. We also provide a dimensionless intensive triad $\widehat{\Psi}$ that enables scale-free comparisons and systemic cut-set averaging. These constructions aim to unify and extend prior pairwise constraints by offering a falsifiable, unit-consistent three-leg threshold under H1–H3.

Roadmap. Section 2 defines the triad and its intensive form; the next section specifies hypotheses (H1–H3). Sections on geometric/entropic/correlation faces motivate the legs; we then derive per-subsystem and systemic inequalities, followed by Hamiltonian-capped and speed-based thresholds. Applications and open-system extensions conclude.

Problem statement. Consider a system on a bounded-degree finite-range interaction graph and any bipartition $(A|\bar{A})$ crossed by at least one interaction. Define the triad $\Pi_A(t) = \sqrt{F_Q(\rho_A; H_A)} S_A I(A : \bar{A})$. Without presupposing positivity of any leg, we obtain that, under H1–H3 and genuine cross-cut coupling, the triad does not vanish in time-average across the cut-set. [24, 1, 3]

2 Quantum Structural Triad: Definition and Role

We define the *quantum structural triad* of a subsystem A in a global state $\rho(t)$ as

$$\Pi_A(t) = \sqrt{F_Q(\rho_A(t); H_A)} S_A(t) I(A : \bar{A}, t), \quad (1)$$

This defines the structural product we bound: local fluctuation sensitivity, local entropy, and cross-cut correlation combined at the subsystem level. *Units:* S_A and I are dimensionless (nats); with time encoding, $\sqrt{F_Q}$ has units of inverse time (s^{-1}), and in the energy form ΔE carries energy (J). Where:

- $F_Q(\rho_A; H_A)$ is the quantum Fisher information (QFI) of the reduced state ρ_A with respect to the local generator H_A , equivalent to the Bures metric curvature at time t , capturing *fluctuation strength*.
- $S_A(t) = -\text{Tr}[\rho_A(t) \log \rho_A(t)]$ is the von Neumann entropy of the subsystem, quantifying *local informational content*.
- $I(A : \bar{A}, t) = S_A(t) + S_{\bar{A}}(t) - S_{A\bar{A}}(t)$ is the quantum mutual information, measuring total (quantum and classical) *correlation* between A and its complement.

Encoding convention. We evaluate $\sqrt{F_Q(\rho_A; H_A)}$ for the unitary encoding generated by H_A on A . For general completely positive trace-preserving (CPTP) trajectories, monotone-metric contractivity gives $\dot{L}_B \leq \frac{1}{2} \sqrt{F_Q}$, used in our speed bounds.

Each leg reflects a distinct facet of quantum structure:

1. $\sqrt{F_Q}$ encodes local dynamical sensitivity to perturbations,
2. S_A encodes intrinsic informational richness,
3. $I(A : \bar{A})$ encodes connectivity to the rest of the system.

We interpret $\Pi_A(t)$ as the *instantaneous structural integrity* of subsystem A . If $\Pi_A(t) = 0$, then A is simultaneously fluctuation-free, pure, or decorrelated, and thus fails to contribute meaningfully to the system’s quantum structure.

Quantum-Structurally Inert (QSI).

Definition 1 (D2 (QSI on an epoch)). *A subsystem A is quantum-structurally inert on a measurable epoch $K \subset [0, \infty)$ with $\delta(K) > 0$ if $\Pi_A(t) = 0$ for almost every $t \in K$.*

Systemic Triad. We define the systemic measure as the sum over all subsystems (rate form):

$$\Pi^{\text{rate}}(t) = \sum_A \Pi_A(t). \quad (2)$$

For brevity we may write $\Pi(t)$ for $\Pi^{\text{rate}}(t)$; when needed we distinguish the *rate form* (Π^{rate}) from the *energy form* (Π^{energy}) introduced later, and we use “Bures angle” with symbol L consistently. Our main result is that $\Pi(t)$, while possibly vanishing at isolated instants, cannot vanish on average across subsystems and over time, *within the stated hypotheses* (H1–H3 and positive-mean conditions), in any locally coupled, physically viable quantum system. This establishes the triad as a *minimal structural threshold for subsystem viability*.

3 Hypotheses and Objects

Conventions. We use $\mathbb{E}[\cdot]$ to denote the invariant/ensemble mean; under ergodicity this equals $\langle \cdot \rangle_{\text{ens}}$.

Status and scope. The structural triad is a hypothesis-qualified proposal. All results are necessary (not sufficient) within H1–H3, finite-range locality, and full-rank epochs (continuity caveat). Extrapolations to emergent geometry lie outside this manuscript and are speculative unless stated otherwise. All logarithms are natural; entropies are measured in nats. For any subsystem A , let d_A denote the Hilbert space dimension of H_A and $d_{\bar{A}}$ that of its complement.

Proved vs Conjectured (Scope). Within H1–H3 and on full-rank epochs, we prove necessary lower bounds for the time average of the triad. The speed based route uses Paley-Zygmund plus ergodic averaging (Birkhoff) to guarantee positive measure active subsets; the QFI-variance relation is used only to translate units (not as a proving step). We do not claim sufficiency. Extensions to algebraic QFT and topological codes are conjectural and placed in the annex; they are not used in proofs.

Table 1: Scope recap: what is proved vs conjectured.

Item	Status	Location
Ergodic positivity (conditional on $\mathbb{E}[\Pi_A] > 0$)	Proved (conditional)	§9.3, §10 (PZ+Birkhoff)
Sufficiency of thresholds	Not claimed	—
Topological/AQFT extension	Conjectural (annex)	A14
QFI variance (unit bridge)	Identity (not used in proof)	§4 Identity 1

Interaction graph and cut-set. We represent the system by a bounded-degree, finite-range interaction graph $G = (V, E)$ associated with the local terms of the Hamiltonian. A cut-set E consists of the nearest-neighbor edges crossing a bipartition $(A|\bar{A})$. Its cardinality is denoted $|E|$, and for intensive averages we use

$$\frac{1}{|E|} \sum_{A \in E} (\cdots). \quad (3)$$

Structural Hypotheses (H1–H3). To state our results precisely, we impose three physically natural hypotheses:

- H1. Locality.** The Hamiltonian is a sum of local terms with finite interaction range and uniformly bounded degree on the interaction graph G .
- H2. Genuine coupling.** Each bipartition defined by the cut-set E includes at least one nonzero coupling term, preventing complete isolation of subsystems.
- H3. Ergodic non-stationarity.** On a full-rank epoch with measure-preserving ergodic dynamics, the triad observable $\Pi_A(t) = \sqrt{F_Q} S_A I(A:\bar{A})$ is bounded (hence in L^2) and no single leg is fixed to zero almost everywhere on that epoch. No assumption of $\mathbb{E}[\Pi_A] > 0$ is made.

Continuity caveat. Rank-change times can accumulate in engineered cases. Our results apply on *full-rank epochs*, where $\sqrt{F_Q}$, S_A , and $I(A : \bar{A})$ are continuous (see Lemma 2).

Remark on H3 and scope of validity. The ergodic formulation of H3 is introduced solely to ensure the correspondence between ensemble and time averages required by the Paley–Zygmund and Birkhoff arguments used in the proof of the structural threshold theorems. This mathematical assumption provides a well-defined measure framework in which the time-averaged triad $\langle \Pi_A \rangle_t$ can be treated as an ensemble expectation, thereby allowing a rigorous nonvanishing lower bound to be derived.

From a physical standpoint, the structural interdependence of energy fluctuations, entropy, and correlations does not rely on ergodicity.

Whenever a subsystem exhibits finite local dynamics, entropy, and coupling, the product $\Pi_A(t) = \sqrt{F_Q(\rho_A; H_A)} S_A(t) I(A : \bar{A}, t)$ remains positive instantaneously. Numerical tests in non-ergodic regimes—integrable, prethermal, and many-body-localized systems (Annex H.16)—confirm that this *structural persistence* survives even when ergodic mixing fails. Ergodicity therefore serves as a mathematical scaffold for the time-average theorem, whereas the persistence of the triad appears as a more general empirical and logical identity of quantum systems.

Dimension bounds. For any bipartition $(A, \bar{A})$, the mutual information is bounded by:

$$I(A : \bar{A}) \leq 2 \log \min(d_A, d_{\bar{A}}). \quad (4)$$

Reference. This standard bound follows from strong subadditivity and basic properties of mutual information [40, 6].

These hypotheses exclude trivial or degenerate regimes (e.g., completely isolated subsystems, frozen dynamics, or rank-changing discontinuities). Within this admissible domain, we obtain that the triad Π_A defines a nonvanishing *structural threshold* for subsystem viability.

Scope Note. Although expressed through subsystem observables, the framework is not subsystem-reductionist. The triad thresholds are intended to characterize the *global* viability of the quantum state, with partitions introduced as analytical tools for structural resolution. We make no ontological claim about subsystems as independently real; they serve as a lens through which system-level structure becomes operationally visible.

Subsystems as analytical lenses. Subsystems here are not ontic units but probes: the objects $\sqrt{F_Q(\rho_A; H_A)}$, S_A , and $I(A : \bar{A})$ are evaluated per cut to resolve structure, while all claims and thresholds are *systemic*, stated as cut-set averages and sums (e.g., $\hat{\Psi}_{\text{cut}}, \Pi(t)$) that diagnose the integrity of the *global* state.

Scope and limitations. All results are stated for finite-dimensional cuts on bounded-degree, finite-range interaction graphs under H1–H3 and on full-rank epochs (continuity caveat). Thresholds are necessary (not sufficient) conditions within these hypotheses. Extensions to algebraic QFT and to topological phases are discussed as outlook; claims there are illustrative rather than established.

4 Geometric Uncertainty: Quantum Speed Limit (QSL) from Bures Angle

The Mandelstam–Tamm form of the quantum speed limit provides a direct link between distinguishability in state space and local energy fluctuations. For a subsystem A , the minimal time τ_A required to evolve by a Bures angle L obeys

$$\tau_A \geq \frac{\hbar L}{\Delta E_A}, \quad (5)$$

where ΔE_A is the time-averaged standard deviation of the local Hamiltonian H_A in the state ρ_A .

Equivalently,

$$\Delta E_A \geq \frac{\hbar L}{\tau_A}. \quad (6)$$

This expresses the *geometric face of uncertainty*: whenever the subsystem undergoes a “distinguishable change” (nonzero Bures angle), local energy fluctuations must be strictly positive. In structural terms, no subsystem can remain indefinitely fluctuation-free while still evolving in Hilbert space. This provides the first leg of the structural triad.

Identity 1 (QFI-Variance Relation). For any density operator ρ and Hermitian generator H ,

$$F_Q(\rho; H) \leq \frac{4}{\hbar^2} \text{Var}_\rho(H), \quad \Delta E \geq \frac{\hbar}{2} \sqrt{F_Q(\rho; H)}.$$

This identity underlies the energy-form rewriting of the triad but is not required for the Paley-Zygmund proof route.

5 Entropic Response: First Law of Entanglement

For small perturbations of a quantum state, the first law of entanglement entropy provides a direct link between entropy variation and modular Hamiltonian expectation values:

$$\delta S_A = \delta \langle H_A^{\text{mod}} \rangle, \quad (7)$$

where H_A^{mod} is the modular Hamiltonian associated with region A .

This relation shows that entropy cannot remain frozen under nontrivial local energy exchange. Whenever the subsystem participates in dynamics, changes in S_A are unavoidable. Entropy thus forms the second leg of the structural triad, encoding the local informational footprint of the subsystem.

Caveat. The first law of entanglement is exact for small variations around a reference state, and is most transparent when modular Hamiltonians are explicitly known (e.g., conformal field theory or holographic settings). For general mixed states, it should be interpreted as a local differential relation rather than a global identity. Nevertheless, it captures the essential fact: local entropy cannot vanish persistently in a coupled, evolving quantum system.

6 Correlation as Intrinsic

The quantum mutual information across a bipartition $(A|\bar{A})$ is defined as

$$I(A : \bar{A}) = S_A + S_{\bar{A}} - S_{A\bar{A}}. \quad (8)$$

It measures the total correlation—both classical and quantum—between the subsystem A and its complement $\bar{A}$.

Entropy is never a purely local quantity: it necessarily reflects correlations with the environment or complementary subsystems. Thus mutual information, as the third leg of the triad, is structurally unavoidable in any nontrivial quantum regime. If correlations vanished identically across a genuine cut, the subsystem would be decoupled and structurally irrelevant.

Structural role. Correlation ensures that local fluctuations and entropy are not isolated artifacts but are embedded in a larger relational structure. Together with $\sqrt{F_Q}$ and S_A , mutual information completes the minimal triad of structural viability.

7 Per-Subsystem Inequality

Combining the fluctuation, entropy, and correlation terms, and using the instantaneous Bures speed bound, we obtain on any active interval:

$$\Delta E_A S_A I(A : \bar{A}) \geq \hbar \dot{L}_B S_A I(A : \bar{A}), \quad (9)$$

where $\dot{L}_B$ is the instantaneous Bures angle rate.

Integrating over time and applying the density-to-time-average lemma on a measurable active set K with lower density $\delta > 0$ and leg-wise floors $\sqrt{F_Q} \geq \varepsilon_F$, $S_A \geq \varepsilon_S$, $I \geq \varepsilon_I$ on K (with Π_A bounded), we arrive at

$$\langle \Pi_A \rangle_t \geq \delta L \frac{S_A I(A : \bar{A})}{\tau_A}. \quad (10)$$

This inequality expresses the *structural inseparability* of the triad. Variance, entropy, and correlation are not expected to collapse simultaneously for a genuinely coupled subsystem within our hypotheses. At least one leg is expected to sustain a strictly positive floor over active epochs, indicating that Π_A does not vanish in time average.

Structural consequence. Whenever $\langle \Pi_A \rangle_t = 0$, the subsystem behaves as quantum-structurally inert: it contributes no persistent fluctuations, entropy, or correlations. The inequality therefore provides a *per-subsystem structural threshold for viability*.

Covariance correction (product vs average). When using floors of the multiplicative triad, one should distinguish $\langle f_1 f_2 f_3 \rangle_t$ from $\prod_i \langle f_i \rangle_t$. In general,

$$\langle f_1 f_2 f_3 \rangle_t = \prod_{i=1}^3 \langle f_i \rangle_t + \sum_{i \neq j} \text{Cov}(f_i, f_j) + \text{Cov}_3(f_1, f_2, f_3), \quad (11)$$

so replacing the time average of the product by the product of time averages is an approximation exact only under weak cross-covariances or narrow distributions. Our thresholds are stated without this replacement; when coarse-grained floors are used, the correction terms can be bounded under mixing/ergodicity assumptions. We do not replace $\langle f_1 f_2 f_3 \rangle$ by $\prod_i \langle f_i \rangle$ in any *stated* bound; Eq. (11) is included as a caution for coarse estimates only.

8 Systemic Inequality

Summing over all subsystems yields the global triad inequality (rate form):

$$\Pi^{\text{rate}}(t) = \sum_A \sqrt{F_Q(\rho_A(t); H_A)} S_A(t) I(A : \bar{A}, t) \geq \sum_A \frac{L S_A I(A : \bar{A})}{\tau_A}. \quad (12)$$

The corresponding *energy* form is

$$\Pi^{\text{energy}}(t) := \sum_A \Delta E_A(t) S_A(t) I(A : \bar{A}, t).$$

As long as the trajectory in state space has nonzero Bures length ($L > 0$), the right-hand side is strictly positive.

Whenever the global state undergoes distinguishable evolution, the triad suggests a *systemic structural threshold*. Even if individual subsystems fluctuate between relevance and irrelevance at specific instants, the ensemble is not expected to sustain a collectively vanishing product across time under the stated assumptions.

Structural consequence. This inequality indicates that, under H1–H3 and genuine coupling, a locally coupled quantum system would be inconsistent with the bound if it remained entirely fluctuationless, unentangled, and pure across all subsystems in time-average. Systemic persistence of the triad thus appears as a generic requirement within these hypotheses for maintaining quantum structural integrity.

9 Intensive Formulation and Root Inequalities

This section introduces the scale-free intensive triad $\widehat{\Psi}$ and states the two structural thresholds (Hamiltonian-capped and speed-based) used in the applications. *Proof roadmap:* the Hamiltonian-capped form follows from intensive normalization and minimal operator norms; the speed-based form combines the Bures speed relation with Paley-Zygmund and Birkhoff (ergodic averaging).

9.1 Normalized Intensive Triad

To compare across systems and isolate structure from scale, we define a dimensionless intensive version of the triad for each cut $(A|\bar{A})$:

$$\widehat{\Psi}_A(t) := \underbrace{\frac{\hbar \sqrt{F_Q(\rho_A(t); H_A)}}{2\|H_A\|_{\text{op}}}}_{\in[0,1]} \cdot \underbrace{\frac{S_A(t)}{\log d_A}}_{\in[0,1]} \cdot \underbrace{\frac{I(A : \bar{A}, t)}{2 \log \min(d_A, d_{\bar{A}})}}_{\in[0,1]}. \quad (13)$$

This rescales Π_A to a dimensionless quantity in $[0, 1]$, enabling comparisons across systems and cuts. Each factor is bounded by standard inequalities, so $0 \leq \widehat{\Psi}_A(t) \leq 1$. We use the cut-set average and its time average:

$$\widehat{\Psi}_{\text{cut}}(t) := \frac{1}{|E|} \sum_{A \in E} \widehat{\Psi}_A(t), \quad \langle \widehat{\Psi}_{\text{cut}} \rangle_t := \liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T \widehat{\Psi}_{\text{cut}}(t) dt.$$

9.2 Hamiltonian-Capped Structural Threshold

Theorem 1 (Hamiltonian-Capped Structural Threshold). *Under H1-H2 (no speed assumptions), for any cut-set E ,*

$$\frac{1}{|E|} \sum_{A \in E} \Delta E_A S_A I(A : \bar{A}) \geq 2 (\log d_{\min})^2 \|H\|_{\min} \langle \widehat{\Psi}_{\text{cut}} \rangle_t, \quad d_{\min} := \min_{A \in E} \min(d_A, d_{\bar{A}}), \quad (14)$$

This threshold is hardware-capped: the floor scales with the minimal local energy norm and minimal subsystem dimension. where $\|H\|_{\min} := \min_{A \in E} \|H_A\|_{\text{op}}$.

Proof sketch. From (13),

$$\sqrt{F_Q} S_A I(A : \bar{A}) := \frac{2\|H_A\|_{\text{op}}}{\hbar} (\log d_A) [2 \log \min(d_A, d_{\bar{A}})] \widehat{\Psi}_A(t).$$

Use $\Delta E_A \geq \frac{\hbar}{2} \sqrt{F_Q}$ to replace $\sqrt{F_Q}$, then average over $A \in E$ and lower-bound $\|H_A\|_{\text{op}}, \log d_A, \log \min(d_A, d_{\bar{A}})$ by their respective minima. Time-averaging yields (14). $\square$

This threshold is *unconditional in speed*: even without assuming a dynamical floor, the normalized structural integrity $\langle \widehat{\Psi}_{\text{cut}} \rangle_t$ implies a positive lower bound set by the *minimal local energy scale* and *minimal subsystem dimension*. It quantifies how structural integrity trades off against hardware scales.

9.3 Speed-Based Structural Threshold

Existence of active times via Paley-Zygmund (no prior active-epoch assumption). By the Paley-Zygmund Epoch Lemma (Lemma 4), any bounded, nonnegative observable in L^2 evolving under a measure-preserving ergodic flow admits, for each $0 < \theta < 1$, a positive-measure set of times K_θ on which it exceeds a fixed fraction of its mean. In particular, for $\Pi_A(t) = \sqrt{F_Q} S_A I(A : \bar{A})$ we obtain $\delta(K_\theta) > 0$ without assuming an “active epoch.”

Theorem 2 (Speed-Based Structural Threshold). *Under H1–H2–H3 and bounded finite-dimensional unitary evolution with $\Pi_A(t) \geq 0$ bounded (so $\Pi_A \in L^2$) and measure-preserving ergodic dynamics, for any $0 < \theta < 1$ and any subsystem A ,*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T \Delta E_A(t) S_A(t) I(A:\bar{A}, t) dt \geq \frac{\hbar}{2} \theta(1-\theta)^2 \frac{(\mathbb{E}[\Pi_A])^3}{\mathbb{E}[\Pi_A^2]}. \quad (15)$$

Choosing $\theta = \frac{1}{2}$ yields the constant prefactor $\hbar/16$.

Note. The bound is conditional: if $\mathbb{E}[\Pi_A] = 0$ the right-hand side vanishes. The choice of $\theta \in (0, 1)$ trades the constant factor $\theta(1-\theta)^2$ against the activation measure; $\theta = \frac{1}{2}$ gives $\hbar/16$.

This threshold is *dynamical*: the Paley-Zygmund Epoch Lemma guarantees a positive-measure set of times where Π_A exceeds a fixed fraction of its mean, and the pointwise bound $\Delta E_A \geq (\hbar/2)\sqrt{F_Q}$ converts this into a strictly positive lower bound for the energy form.

Relation to prior QFI–entropy bounds. The speed-based inequality above connects directly to the family of quantum-speed-limit results established by Deffner and Lutz [14], who proved that the quantum Fisher information F_Q defines the geometric speed of state evolution through the Bures metric. Their bound,

$$\tau_{\text{QSL}} = \frac{\sin^2[\mathcal{L}(\rho_0, \rho_\tau)]}{\frac{1}{\tau} \int_0^\tau \sqrt{F_Q(\rho_t; H_t)} dt}, \quad (16)$$

identifies $\sqrt{F_Q}/2$ as the maximal instantaneous speed of change of a quantum state. In the present formulation, this temporal leg $\sqrt{F_Q}$ is coupled to the entropic and correlational factors S and $I(A:\bar{A})$, producing the composite structural invariant $\Pi_A = \sqrt{F_Q} S I$. Hence, Theorem 2 generalizes the Deffner–Lutz quantum-speed-limit inequality from a purely dynamical constraint to a *structural persistence* condition that remains positive whenever all three legs of the triad admit finite floors.

Note. We do not rederive Deffner–Lutz or related QSL results; we only couple their speed leg to entropic and correlational legs in a new product threshold.

Relation to thermodynamic and information-theoretic bounds. The present speed-based inequality complements the family of entropy-production bounds established in quantum thermodynamics. In open or driven systems, the rate of entropy change obeys Spohn’s inequality

$$\dot{S}_{\text{prod}} \geq 0, \quad (17)$$

and the generalized quantum speed limits of Deffner and Lutz [14] and Liu *et al.* [15] connect this rate to the quantum Fisher information and the Bures distance. Our formulation integrates these relations by coupling the dynamical leg $\sqrt{F_Q}$ to both the entropic leg S_A and the correlational leg $I(A:\bar{A})$, yielding a joint, unit-consistent product threshold. Thus the triad inequality extends existing QFI–entropy and entropy-production bounds from pairwise to fully three-way structural form.

9.4 Trade-offs and Scaling

Both thresholds quantify *trade-offs*: if one leg (fluctuation, entropy, or correlation) is small, the others must compensate to keep the left-hand side above threshold. System-size enters via $(\log d_{\min})^2$: larger minimal Hilbert spaces demand stronger combined resources to maintain structural integrity.

10 Technical Lemmas

The following lemmas establish the auxiliary bounds needed to derive the structural thresholds.

Lemma 1 (Instantaneous Bures speed). *For unitary dynamics generated by H_A ,*

$$\dot{L}_B(t) = \frac{1}{2} \sqrt{F_Q(\rho_A(t); H_A)}. \quad (18)$$

For general CPTP dynamics, $\dot{L}_B(t) \leq \frac{1}{2} \sqrt{F_Q}$.

Lemma 2 (Continuity and full-rank epochs). *For finite-dimensional piecewise C^1 trajectories, the functions $t \mapsto S_A(t)$, $t \mapsto I(A : \bar{A}, t)$, and $t \mapsto \sqrt{F_Q(\rho_A(t); H_A)}$ are continuous on intervals where $\rho_A(t)$ is full rank. Rank-change times form a set of measure zero.*

Definition 2 (D1 (Lower density)). *For measurable $K \subseteq [0, \infty)$, define*

$$\delta(K) := \liminf_{T \rightarrow \infty} \frac{|K \cap [0, T]|}{T}. \quad (19)$$

Lemma 3 (Density-to-time-average). *Let $f : [0, \infty) \rightarrow [0, \infty)$ be measurable, and suppose $f(t) \geq \alpha > 0$ for all $t \in K$ with $\delta(K) > 0$. Then*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(t) dt \geq \delta(K) \alpha. \quad (20)$$

Lemma 4 (Paley-Zygmund Epoch Lemma). *Let $\Pi_A(t) = \sqrt{F_Q(\rho_A; H_A)} S_A(t) I(A : \bar{A}, t) \geq 0$ be bounded and evolve under measure-preserving ergodic dynamics on a finite-dimensional space. Then for any $0 < \theta < 1$,*

$$\mu\{t : \Pi_A(t) \geq \theta \mathbb{E}[\Pi_A]\} \geq (1 - \theta)^2 \frac{(\mathbb{E}[\Pi_A])^2}{\mathbb{E}[\Pi_A^2]} > 0,$$

Corollary (PZ time-average). *Consequently,*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T \Pi_A(t) dt \geq \theta(1 - \theta)^2 \frac{(\mathbb{E}[\Pi_A])^3}{\mathbb{E}[\Pi_A^2]}.$$

Remark. *The bound is conditional: if $\mathbb{E}[\Pi_A] = 0$ the righthand side vanishes; when $\mathbb{E}[\Pi_A] > 0$ it yields a strictly positive floor. Proof. Apply Paley-Zygmund to the nonnegative L^2 function Π_A and identify time measure with invariant measure via the Birkhoff Ergodic Theorem. The statement holds on each ergodic component; for non-ergodic systems it applies componentwise.* $\square$

Structural Lemmas (L1–L4).

Lemma 5 (L1: Reduced motion from global motion). *Under H1–H2–H3, reduced dynamics is generically non-stationary for almost every time in K except possibly a null set. Symmetry-protected subsystems can remain stationary despite global motion; our systemic bounds aggregate over cuts or apply on subsets without such symmetries.*

Lemma 6 (L2: Fluctuation leg). *From non-stationarity on K and Lemma 1, there exists a subset $K_F \subset K$ with $\delta(K_F) > 0$ and $\sqrt{F_Q(\rho_A; H_A)} > 0$ for almost every $t \in K_F$.*

Lemma 7 (L3: Correlation leg). *With a cross-cut coupling (H2) and locality (H1), Lieb–Robinson propagation implies $I(A : \bar{A}) > 0$ after a microscopic time on a positive-measure subset $K_I \subset K$.*

Lemma 8 (L4: Entropy leg). *For closed pure global states, $I = 2S_A$. For mixed/open dynamics, Spohn’s inequality implies strictly positive local entropy production away from steady state on a positive-measure subset $K_S \subset K$.*

11 Discussion: Structural Relevance and Irrelevance

We interpret the bounds, quantify trade-offs among legs, and outline limits of use (floors estimation, finite-size effects, and symmetry caveats), connecting the inequalities to practical design guidance.

Witnessable floors and active epoch. In practice, floors $(\varepsilon_F, \varepsilon_S, \varepsilon_I)$ can be estimated from time-series using conservative quantiles on the active epoch $K = \{t : \sqrt{F_Q} \geq \varepsilon_F, S_A \geq \varepsilon_S, I \geq \varepsilon_I\}$ and $\delta(K)$ as the fraction of frames in K (optionally with rolling or lag tolerance). When direct QFI is costly, scalable witnesses (e.g., commutator ladders, variance caps, Bures-angle sampling) can be used to bound $\sqrt{F_Q}$ from below, yielding conservative $\text{RHS}_{\text{speed}}$ without altering the theory.

11.1 Complementary Thresholds

The two inequalities derived above provide complementary perspectives on subsystem viability:

- **Hamiltonian-capped threshold:** unconditional, tied to the smallest operator norm $\|H\|_{\min}$ and subsystem dimension $d_{\min}$. It reflects the *structural potential* inherent in the system’s composition.
- **Speed-based threshold:** conditional on a nonzero Bures/QFI speed floor, it converts dynamical activity into a strictly positive structural lower bound, typically tighter in nonequilibrium regimes.

These represent two physically consistent, unit-consistent routes to nonvanishing lower bounds on the fluctuation–entropy–correlation product.

11.2 Units and Physical Meaning

The structural triad combines distinct units:

- $\sqrt{F_Q}$ carries units of inverse time (distinguishability rate),
- ΔE carries energy,
- S and I are dimensionless (nats).

The Hamiltonian-capped threshold introduces an energy scale via $\|H\|_{\min}$, while the speed-based threshold introduces a temporal scale via the Bures speed. In both cases, the inequality implies that the product of these resources cannot vanish under admissible conditions.

11.3 Boundary Cases and Structural Irrelevance

The thresholds become trivial precisely when an *excluded hypothesis* holds:

1. $I \equiv 0$ (violates H2: no genuine cross-cut coupling),
2. $\sqrt{F_Q} \equiv 0$ or zero-speed protocols (violates H3: global non-stationarity),
3. Rank-changing discontinuities beyond measure-zero sets (covered by continuity caveat).

In each case, the subsystem behaves as *quantum-structurally inert* (QSI): incapable of acting as a reliable mediator, memory, or participant in nontrivial dynamics.

11.4 Trade-offs and Design Rules

The inequalities highlight explicit trade-offs:

- Small entropy can be compensated by larger fluctuations or correlations,
- Weak correlations demand stronger fluctuations and entropy,
- Minimal subsystem dimension $d_{\min}$ sets a rising structural cost with scale.

Thus subsystem viability is governed not by any single resource but by their interdependence. This suggests practical design rules: viable subsystems must maintain all three legs above thresholds in some combination, and resource deficits in one leg require reinforcement in the others.

11.5 Experimental Accessibility

Few-body experiments can probe the triad directly via:

- QFI estimates from Bures angle trajectories,
- Entropy via tomography or entropic witnesses,
- Mutual information via correlation measurements.

Many-body systems require witness-based or coarse estimators, but our finite-window QFI bound (Annex; see § Annex A15) can enable positive certification even with partial data.

Corollary (no admissible QSI epochs)

If H1–H2–H3 hold, then A is not expected to be QSI on K : otherwise $\Pi_A = 0$ almost everywhere on a positive-density epoch K^* would contradict the thresholds above.

12 Quantum Computing Applications

How to read the thresholds: estimate floors $(\varepsilon_F, \varepsilon_S, \varepsilon_I)$ and the active epoch density $\delta(K)$, then certify structural viability when $\Pi \geq \max\{\text{RHS}_{\text{cap}}, \text{RHS}_{\text{speed}}\}$.

The structural thresholds provide practical, physics-based exclusion criteria for quantum technologies in this framework. Because they are *necessary conditions* within H1–H3, subsystems persistently below threshold are unlikely to exhibit sufficient quantum functionality, supporting the exclusion of nonviable qubits, gates, or memories in a resource-aware workflow.

12.1 Qubit Viability Certification

For each physical qubit i , measure or estimate

$$\Pi_i = \sqrt{F_Q(\rho_i; H_i) S_i I(i : \bar{i})}. \quad (21)$$

A qubit is structurally viable, within estimator uncertainty, if

$$\Pi_i \geq \max\{\text{Hamiltonian-capped threshold}, \text{Speed-based threshold}\}.$$

Qubits with Π_i below threshold are quantum-structurally inert in this framework and are unlikely to support reliable quantum logic under the stated assumptions.

12.2 Error Correction Prioritization

Error correction is resource-intensive. Our framework helps identify qubits worth correcting:

$$\text{Apply QEC only if } \Pi_i \geq \text{threshold}.$$

When Π_i remains below threshold, additional error correction is unlikely to be effective within this criterion, guiding resource allocation in large-scale devices.

12.3 Quantum Memory Lifetime

The speed-based inequality provides a bound on memory coherence:

$$\text{Quantum memory may be viable while } \delta(t) \varepsilon_F(t) \varepsilon_S(t) \varepsilon_I(t) \geq \frac{2}{\hbar} \Pi_{\min}. \quad (22)$$

When any component falls below its floor, preserving quantum coherence becomes unlikely within this criterion.

12.4 Gate Operation Certification

For multi-qubit gates, structural viability is favored when

$$\Pi_{\text{gate}} \geq \frac{\hbar}{2} \delta_{\text{gate}} \varepsilon_F \varepsilon_S \varepsilon_I \quad (23)$$

throughout the gate execution. Gates failing this criterion are unlikely to realize genuine quantum behavior, even when classical fidelity measures appear high.

12.5 NISQ Device Optimization

In noisy intermediate-scale devices, the thresholds enable:

- identification of qubits that are “quantum enough” to include in algorithms,
- bounding of maximum circuit depth before triad decay below threshold,
- optimization of connectivity patterns based on correlation persistence.

12.6 Towards Principled Design

Current practice in quantum engineering is heuristic, guided by coherence times and error rates. Our framework introduces principled design rules grounded in first-principles physics. It answers questions such as:

- “Is this qubit structurally capable of quantum computation?”
- “When should error correction be abandoned?”
- “What are the fundamental limits of this processor architecture?”

By providing definitive exclusion criteria, the structural triad transforms quantum engineering from empirical tuning to threshold-driven design.

12.7 Toy Example: Present-day qubit estimate

Setup. Consider a superconducting qubit with frequency $f = 5$ GHz, $T_1 \approx 100 \mu\text{s}$, and a conservative $T_2 \approx 80 \mu\text{s}$. Take a representative local entropy $S \approx 1$ nat and mutual information $I \approx 0.1$ (small but nonzero cross-cut correlation), active-epoch density $\delta \approx 0.3$, subsystem dimensions $d_A = 2$, $d_{\min} = \min(d_A, d_{\bar{A}}) = 2$. We estimate a speed floor by $\varepsilon_F \approx 1/T_2$ and use $\|H_A\|_{\text{op}} \approx \hbar\omega/2$ with $\omega = 2\pi f$ for the intensive normalization.

Results. With these choices we obtain

- $\omega \approx 3.14 \times 10^{10} \text{ s}^{-1}$, $\varepsilon_F \approx 1.25 \times 10^4 \text{ s}^{-1}$,
- product rate $\Pi_{\text{rate}} \approx \varepsilon_F S I \approx 1.25 \times 10^3 \text{ s}^{-1}$,
- speed-based energy threshold $\text{RHS}_{\text{speed}} = (\hbar/2) \delta \varepsilon_F S I \approx 2.0 \times 10^{-32} \text{ J}$,
- intensive triad $\hat{\Psi}$ (with the above normalization) is small but positive ($\approx 4.1 \times 10^{-8}$), consistent with a weakly correlated single-qubit cut and strong hardware normalization.

These values illustrate how to form a conservative certification using present-day parameters: pick small but nonzero S and I consistent with device telemetry, a floor for $\sqrt{F_Q}$ via $1/T_2$, and a modest active-epoch density δ . The product rate Π_{rate} and the speed-based threshold scale linearly with these choices, providing a transparent knob-level interpretation. *Remark.* These estimates are illustrative (small- I regime) and device-dependent; quantities are reported in SI units for transparency.

13 Open-System Extension: Structural Relevance Under Dissipation

13.1 Setting and Hypotheses

We extend the structural triad to dissipative dynamics governed by a local Lindbladian

$$\mathcal{L} = \sum_X \mathcal{L}_X,$$

with finite interaction range and bounded degree. The CPTP propagator is $\Lambda_t = e^{t\mathcal{L}}$ (or time-periodic for driven steady states). Cuts are defined by the same nearest-neighbor cut-set E . Each cut $(A|\bar{A})$ is assumed genuinely coupled by at least one Lindblad or Hamiltonian term. Dissipative Lieb–Robinson bounds ensure finite-speed propagation of influence.

H3 (open). There exists a measurable epoch $K \subset [0, \infty)$ of non-stationary CPTP evolution (away from a fixed point) with $\delta(K) > 0$.

13.2 Open-System Triad Reformulation

Even under dissipation, the triadic structure persists in generalized form:

- **Fluctuations:** The QFI leg is replaced by a channel speed measure F_{chan} , defined via monotone Riemannian metrics on CPTP maps [28].
- **Entropy:** The entropy leg becomes the local entropy production rate

$$\dot{S}_{\text{prod},A}(t) := -\frac{d}{dt} D(\rho_A(t) \| \sigma_A) \geq 0,$$

where σ_A is the full-rank local steady state (Spohn's inequality), strictly > 0 away from steady state.

- **Correlations:** The correlation leg is extended to system–environment mutual information, accessible via Stinespring dilations.

13.3 Intensive Open-System Triad

Define the normalized intensive open-system triad:

$$\hat{\Psi}_A^{\text{open}}(t) := \underbrace{\frac{\sqrt{F_{\text{chan}}(\Lambda_t|A)}}{\sqrt{F_{\text{max}}}}}_{\in[0,1]} \cdot \underbrace{\frac{\dot{S}_{\text{prod},A}(t)}{\dot{S}_{\text{max}}}}_{\in[0,1]} \cdot \underbrace{\frac{I(\text{sys} : \text{rest}, t)}{2 \log d_{\text{min}}}}_{\in[0,1]}. \quad (24)$$

The cut-set intensive average is

$$\hat{\Psi}_{\text{cut}}^{\text{open}}(t) = \frac{1}{|E|} \sum_{A \in E} \hat{\Psi}_A^{\text{open}}(t), \quad \overline{\hat{\Psi}_{\text{cut}}^{\text{open}}} := \liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T \hat{\Psi}_{\text{cut}}^{\text{open}}(t) dt.$$

Theorem 3 (Open-System Structural Threshold). *Under locality, genuine coupling, finite-speed dissipative propagation, and existence of a non-stationary epoch K with density $\delta > 0$ and leg-wise positivity away from steady state,*

$$\overline{\hat{\Psi}_{\text{cut}}^{\text{open}}} \geq \delta \frac{\varepsilon_F}{\sqrt{F_{\text{max}}}} \cdot \frac{\varepsilon_S}{\dot{S}_{\text{max}}} \cdot \frac{\varepsilon_I}{2 \log d_{\text{min}}} > 0. \quad (25)$$

Even with dissipation, subsystems are unlikely to persist in a state of structural irrelevance except at true fixed points. Nontrivial open-system evolution tends to sustain nonvanishing fluctuation, entropy production, and correlations, maintaining quantum structural integrity in the presence of noise and dissipation, within the stated hypotheses.

14 Conclusions

Under the stated hypotheses H1–H3—locality, genuine cross-cut coupling, and ergodic non-stationarity—the triadic observable

$$\Pi_A(t) = \sqrt{F_Q(\rho_A; H_A)} S_A I(A; \bar{A})$$

cannot vanish in time average, provided that its ensemble mean is positive. By the *Ergodic Positivity Lemma* (Paley–Zygmund + Birkhoff), if $\mathbb{E}[\Pi_A] > 0$ then for any $0 < \theta < 1$ the bounded observable satisfies

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T \Pi_A(t) dt \geq \theta(1 - \theta)^2 \frac{(\mathbb{E}[\Pi_A])^3}{\mathbb{E}[\Pi_A^2]} > 0.$$

In particular, choosing $\theta = \frac{1}{2}$ yields a prefactor $\frac{1}{8}$. This result is therefore proved as a *conditional property* under H1–H3 and $\mathbb{E}[\Pi_A] > 0$; establishing the positivity of the ensemble mean itself under H1–H2 and non-stationarity is left as a conjecture supported by numerical and phenomenological evidence.

The triad expresses a minimal condition for subsystem viability: a quantum system cannot retain dynamical relevance while all three components—fluctuation, entropy, and correlation—vanish simultaneously in time average. From locality and non-stationarity we obtained two complementary quantitative thresholds:

- a *Hamiltonian-capped inequality*, governed by minimal local energy scales and subsystem dimensions;
- a *speed-based inequality*, governed by Bures/QFI evolution rates and entropic/correlation floors.

Together, these bounds indicate that the three quantities are structurally interdependent: a persistent deficit in one component must be balanced by the others, while the simultaneous vanishing of all three marks structural inertness under the stated hypotheses. This inseparability delineates a threshold between inert mathematical possibility and physically participatory behaviour.

Although the framework is formulated in terms of subsystem observables—reduced fluctuations, local entropy, and mutual information—it is not reductionist. Subsystems serve as analytical constructs introduced to resolve structure within the total state. The theory is fundamentally systemic: it characterizes the integrity of the full quantum state, employing partitions as diagnostic tools rather than ontological divisions.

Subsystems falling below the derived thresholds are structurally inert in this sense: unlikely to store information, mediate interactions, or sustain distinguishable evolution. In contrast, persistent triadic activity signals the minimal conditions for quantum persistence. This exclusion principle applies to pure and mixed states alike and extends naturally to Lindbladian open systems through entropy production.

Scope and outlook. Possible extensions to algebraic or topological regimes are heuristic and summarised separately in Annex F; they are not part of the proved results. Future work will refine computational annexes, develop experimental witnesses, and explore connections between subsystem structure, modular dynamics, and emergent geometry. Through this lens, quantum viability serves as a working organising principle of physical structure.

Structural persistence and plateau regimes. In many-body systems, long plateau intervals may arise from quasi-integrable or prethermal sectors. These modify the temporal distribution of activity but not its existence. Within each invariant component, whenever $\mathbb{E}[\Pi_A] > 0$, the Paley–Zygmund+Birkhoff route guarantees a positive-measure active subset, ensuring a nonvanishing time-average of the triad. Even when global ergodicity breaks down or mixing is slow, the structural product $\Pi_A(t)$ remains persistently positive, as confirmed by non-ergodic tests (Annex H.16). Ergodicity thus serves only as a technical bridge between ensemble and time averages, while the underlying structural persistence of the triad reflects a broader physical identity that extends across ergodic and non-ergodic regimes alike.

On falsifiability and present limitations

In the Popperian sense of testability, the triadic principle presently has a *dual status*. Within the hypotheses H1–H3 its *time-average non-vanishing* follows conditionally from the Paley–Zygmund + Birkhoff route, provided that the ensemble mean $\mathbb{E}[\Pi_A] > 0$. Beyond these assumptions, its claimed universality is advanced as a *structural postulate*: a conjectural statement about what it means for a quantum system to sustain viable dynamics.

A *genuine falsification* would require a controllable quantum platform in which the three constituent quantities—local energy fluctuation (quantum Fisher information), von Neumann entropy, and mutual information—*all vanish simultaneously* while the system remains dynamically functional, i.e. still capable of evolving, storing information, or mediating interactions under its own dynamics.

Such a configuration would conflict with established monotonicity relations of quantum mechanics: the *energy-time uncertainty bound* [25], *entropy-production positivity* [12], and *entanglement-generation monotonicity* implied by strong subadditivity [26]. Hence the triadic constraint appears to delineate the physical boundary of admissible dynamics *within standard quantum mechanics*.

Because the triad is multiplicative, the vanishing of any single leg immediately nullifies the product, rendering the subsystem structurally inert even if the other components remain finite. Partial vanishing (one or two legs) therefore signals inertness and does not falsify the conjecture; only *simultaneous* vanishing under sustained evolution would.

To date, every realizable system examined in the numerical annexes—including TEBD and eigenstate tests (Annex A1–A15)—preserves a finite triadic product. Nevertheless, the conjecture of universal interdependence must be regarded as *open* until either a formal proof or an explicit counterexample is produced. As Popper observed, “the criterion of the scientific status of a theory is its falsifiability, or refutability, or testability” [27]; the triadic framework satisfies this criterion precisely, though its ultimate universality remains to be established.

In future work we will explore whether the present conditional theorem can be reformulated within a fully algebraic setting, where the non-vanishing of the triad might follow from the structural properties of faithful quantum states and non-central observables. Such a modular or operator-theoretic reformulation, if achievable, would remove explicit dynamical assumptions and could reveal the triadic condition as an intrinsic feature of quantum structure.

Acknowledgements

This project began a long time ago, with a long-standing question: how does the principle *natura abhorret vacuum*—the idea that physical reality never admits absolute emptiness—retain its validity from the microscopic to the cosmological scale? To explore that question I carried out a symbolic reduction of representative equations drawn from quantum, thermodynamic, relativistic, and cosmological theories. When all non-universal or model-specific quantities were removed, three elements remained invariant across those domains: energy fluctuations, entropy, and Planck’s constant. In pure states, entropy manifests as entanglement entropy, revealing the same structure in its quantum form. These invariants became the foundation of the structural triad developed and verified in this work.

Large-language-model systems were employed as analytical and computational assistants. To limit both human and algorithmic bias and to control potential hallucination, all theoretical results were subsequently verified with the *Rocq* proof assistant and cross-validated through independent numerical simulations.

The full verification materials, including Rocq proof scripts, computational notebooks, and simulation outputs, are available from the author’s GitHub repository upon request.

Although not a professional physicist or formally affiliated researcher, I carried out the conceptual development, interpretation, and synthesis independently. The involvement of AI and proof-assistant tools should be understood as a form of assisted reasoning, not as authorship. I retain complete epistemic responsibility for the arguments, methods, and conclusions advanced here and accept responsibility for any remaining errors.

Annexes: Computational and Formal Backing

The annexes compile notation and normalization details, auxiliary lemmas and proofs, and the full reproducibility package: computational validations A1–A15 alongside the Rocq/Coq verification. For navigation, see the annex index at Section H.

A Notation, Units, and Normalizations

We recall the subsystem triad (Eq. (1))

$$\Pi_A(t) = \sqrt{F_Q(\rho_A(t); H_A)} S_A(t) I(A:\bar{A}, t),$$

and its intensive normalized form (Eq. (13))

$$\hat{\Psi}_A(t) ::= \frac{\hbar \sqrt{F_Q(\rho_A; H_A)}}{2 \|H_A\|_{\text{op}}} \cdot \frac{S_A}{\log d_A} \cdot \frac{I(A:\bar{A})}{2 \log \min(d_A, d_{\bar{A}})} \in [0, 1].$$

Units. With time encoding, $\sqrt{F_Q}$ has units of inverse time, ΔE has units of energy, while S and I are dimensionless (nats). The Hamiltonian-capped inequality introduces an energy scale via $\|H_A\|_{\text{op}}$, and the speed-based form introduces a temporal scale via Bures/QFI speed.

Normalization bounds. (i) $\hbar\sqrt{F_Q}/(2\|H_A\|_{\text{op}}) \leq 1$ (unitary time encoding). (ii) $S_A \leq \log d_A$. (iii) $I(A:\bar{A}) \leq 2 \log \min(d_A, d_{\bar{A}})$. Hence $\hat{\Psi}_A \in [0, 1]$.

B Proof Details for Auxiliary Lemmas

Lemma 9 (Instantaneous Bures speed; Lemma 1). *For unitary dynamics on A , $\dot{L}_B(t) = \frac{1}{2}\sqrt{F_Q(\rho_A(t); H_A)}$. For general CPTP dynamics, $\dot{L}_B \leq \frac{1}{2}\sqrt{F_Q}$.*

Proof. The Bures angle rate equals half the SLD metric speed [3]. For unitary encoding this is an equality; for CPTP maps, contractivity of monotone metrics implies an upper bound. $\square$

Lemma 10 (Continuity and full-rank epochs; Lemma 2). *On intervals where $\rho_A(t)$ is full rank and piecewise C^1 , the maps $t \mapsto S_A(t)$, $t \mapsto I(A:\bar{A}, t)$, and $t \mapsto \sqrt{F_Q}$ are continuous; rank-change times have measure zero.*

Proof. Spectral continuity of finite-dimensional density matrices implies continuity of von Neumann entropy away from eigenvalue collisions; mutual information is a linear combination of entropies. Standard perturbation theory and continuity of the SLD metric on full-rank states yield continuity of $\sqrt{F_Q}$. Rank changes occur at isolated times in piecewise-smooth trajectories, forming a null set. $\square$

Lemma 11 (Density-to-time-average; Lemma 3). *If $f \geq 0$ and $f(t) \geq \alpha > 0$ for all $t \in K$ with lower density $\delta(K) > 0$, then $\liminf_{T \rightarrow \infty} \frac{1}{T} \int_0^T f dt \geq \delta(K) \alpha$.*

Proof. For any $T > 0$, $\int_0^T f dt \geq \int_{K \cap [0, T]} f dt \geq \alpha |K \cap [0, T]|$. Divide by T and take $\liminf$ to obtain the claim. $\square$

Structural Lemmas L1–L4. Under locality (H1) and genuine cross-cut coupling (H2), Lieb–Robinson bounds imply finite-speed propagation and, after a microscopic time, strictly positive mutual information across a genuine cut on a positive-measure subset (L3) [24]. Global non-stationarity (H3) excludes protocols with vanishing Bures speed on a positive-density epoch, ensuring a positive-speed subset for $\sqrt{F_Q}$ (L2). For closed pure states, $I = 2S_A$ implies an entropy floor whenever correlations are present; for open systems, Spohn’s inequality yields strictly positive local entropy production away from steady state (L4) [12]. Reduced non-stationarity (L1) follows generically from global motion except possibly on a null set.

C Proofs of the Main Inequalities

Theorem (Hamiltonian-Capped Structural Threshold; Thm. 1). *For any cut-set E under H1–H2,*

$$\frac{1}{|E|} \sum_{A \in E} \Delta E_A S_A I(A:\bar{A}) \geq 2(\log d_{\min})^2 \|H\|_{\min} \langle \hat{\Psi}_{\text{cut}} \rangle_t.$$

Proof. From Eq. (13), $\sqrt{F_Q} S_A I = \frac{2\|H_A\|_{\text{op}}}{\hbar} (\log d_A) [2 \log \min(d_A, d_{\bar{A}})] \hat{\Psi}_A$. Using the inequality $\Delta E_A \geq (\hbar/2)\sqrt{F_Q}$ (Identity 4) we obtain the corresponding energy form. Hence

$$\Delta E_A S_A I \geq \|H_A\|_{\text{op}} (\log d_A) [2 \log \min(d_A, d_{\bar{A}})] \hat{\Psi}_A.$$

Averaging over $A \in E$ and lower-bounding by $\|H\|_{\min}$, $\log d_{\min}$ yields the claim after time averaging. $\square$

Theorem (Speed-Based Structural Threshold; Thm. 2). *Under H1–H2–H3, there exist $\varepsilon_F, \varepsilon_S, \varepsilon_I > 0$ and $K^* \subset K$ with $\delta(K^*) > 0$ such that*

$$\frac{1}{|E|} \sum_{A \in E} \Delta E_A S_A I(A; \bar{A}) \geq \frac{\hbar}{2} \delta(K^*) \varepsilon_F \varepsilon_S \varepsilon_I.$$

Proof. On K^* , the leg-wise floors hold almost everywhere. Using the inequality $\Delta E_A \geq (\hbar/2)\sqrt{F_Q}$ yields pointwise $\Delta E_A S_A I \geq (\hbar/2)\varepsilon_F \varepsilon_S \varepsilon_I$. Apply Lemma 3 to the nonnegative function $f(t) = (1/|E|) \sum_A \Delta E_A S_A I$ with $K = K^*$. $\square$

D Cut-Set Averaging and Scaling

Let E be the set of nearest-neighbor edges crossing a bipartition. The intensive average $\hat{\Psi}_{\text{cut}}(t) = \frac{1}{|E|} \sum_{A \in E} \hat{\Psi}_A(t)$ is dimensionless and $\in [0, 1]$. The Hamiltonian-capped inequality scales with $(\log d_{\min})^2 \|H\|_{\min}$, making explicit how minimal subsystem dimension and minimal local energy scale set the structural cost of viability. For lattice systems with fixed on-site dimension, the factor grows with the size of the minimal balanced cut, matching the observed growth of raw Π and the stabilization of intensive measures recorded in the simulations.

E Open-System Extension: Details

For local Lindbladians $\mathcal{L} = \sum_X \mathcal{L}_X$ with finite range and bounded degree, define a channel speed leg F_{chan} via a contractive monotone Riemannian metric [28] (e.g., the Bures/SLD metric realized by the pullback from the Jamiołkowski state). Let

$$\hat{\Psi}_A^{\text{open}} ::= \frac{\sqrt{F_{\text{chan}}(\Lambda_t|A)}}{\sqrt{F_{\text{max}}}} \cdot \frac{\dot{S}_{\text{prod},A}}{\dot{S}_{\text{max}}} \cdot \frac{I(\text{sys:rest})}{2 \log d_{\min}} \in [0, 1].$$

Spohn’s inequality [12] yields $\dot{S}_{\text{prod},A} \geq 0$, strictly > 0 away from steady state. Dissipative Lieb–Robinson bounds provide finite-speed influence. With leg-wise positivity on a subset K of density $\delta > 0$, the time-average of the cut-set intensive measure obeys Theorem 3 as stated in the main text.

F Conjectural Extensions — Topological and Algebraic Directions

The remarks below are heuristic and outline potential generalizations of the structural triad to topological codes and algebraic quantum field theory. No formal derivations are attempted; these notes indicate directions for future work.

The structural triad, as derived above, presupposes locality and dynamical non-stationarity (H1–H3). However, certain quantum systems—notably the toric code—exhibit structural persistence in the absence of both. This motivates a conjectural extension of the framework to globally topological, non-dynamical quantum orders.

Mathematical Structure

The toric code Hamiltonian on an $L \times L$ torus,

$$H = -J_e \sum_v A_v - J_m \sum_p B_p, \tag{26}$$

acts on $N = 2L^2$ qubits located on the edges of the lattice. Here $A_v = \prod_{i \in v} \sigma_i^x$ and $B_p = \prod_{i \in p} \sigma_i^z$ are the mutually commuting star and plaquette stabilizers. Despite its static ground-space structure, the toric code retains finite entropy, mutual information, and logical fluctuations under nonlocal observables.

Ground-space structure. All stabilizers commute and satisfy $A_v^2 = B_p^2 = I$, yielding

$$\dim \mathcal{H}_{\text{GS}} = 4, \quad (27)$$

with logical operators W_x, W_z forming noncontractible Wilson loops that anticommute on the torus.

Logical fluctuations. For the mixed ground-state density operator,

$$\rho_{\text{mix}} = \frac{1}{4} \sum_{i=1}^4 |\psi_i\rangle\langle\psi_i|, \quad (28)$$

the logical Wilson loops satisfy

$$\langle W_x \rangle = \langle W_z \rangle = 0, \quad \Delta W_x = \Delta W_z = 1, \quad (29)$$

indicating persistent logical uncertainty even in the absence of dynamical evolution.

Entropic and correlational structure. For half-torus bipartitions at $L = 2$,

$$S_A \approx 2.77 \text{ nats}, \quad (30)$$

$$I(A : \bar{A}) \approx 4.16, \quad (31)$$

leading to the observed topological triad values

$$\Pi_{x,z}^{\text{topo}} = \Delta W_{x,z} S_A I(A : \bar{A}) \approx 11.53. \quad (32)$$

The equality $\Pi_x^{\text{topo}} = \Pi_z^{\text{topo}}$ reflects isotropy under $J_e = J_m$.

Interpretation

Even without dynamical evolution or local coupling, the triad remains finite. The logical fluctuations $\Delta W_{x,z}$ substitute for local energy variance, while S_A and $I(A : \bar{A})$ preserve their informational roles. The toric code therefore appears to satisfy a generalized static inequality,

$$\Pi^{\text{topo}} = \Delta_{\text{logical}} S_A I(A : \bar{A}) > 0, \quad (33)$$

expressing the structural viability of topological quantum systems.

Implications

- The persistence of $\Pi^{\text{topo}} > 0$ indicates that systemic viability can be maintained through global logical structure, not only through dynamical coupling.
- Logical operators act as “virtual dynamics,” sustaining distinguishability and informational curvature in a stationary code space.
- This extension suggests that the triad characterizes *systemic structural persistence*—the capacity of a quantum system to remain informationally active—whether sustained by local motion or by topological coherence.

Future Computations

To formalize this conjecture, subsequent work will:

- Extend simulations to $L = 3, 4$ using stabilizer algebra rather than dense diagonalization;
- Evaluate the scaling of S_A , $I(A : \bar{A})$, and Π^{topo} with system size L ;
- Test whether Π^{topo}/L^2 approaches a constant, defining a topological invariant of structural integrity.

This conjectural extension positions the triad as a universal descriptor of quantum structural persistence, valid across both local and topological phases.

F Formal Verification (Rocq/Coq)

The derivations underlying the structural thresholds have been mechanized in the Rocq/Coq development under `Hyppopotamus/rocq-project`. The formalization mirrors the paper’s structure (notation, normalization, product lifts) and includes the modular triad architecture (system interface, leg floors, normalization, sum lifting). Build and checking use the generated Coq makefiles.

Build. From the repository root (build inside the clean tree):

```
make -C Hyppopotamus/rocq-project/src_clean -f CoqMakefile -j"$(nproc)" | cat
```

Logical checking (coqchk). Core check (recommended):

```
# Legacy constructive + triad variant
coqchk -R Hyppopotamus/rocq-project/src_clean ESSEClean \
  ESSEClean.ESSEBoxed ESSEClean.ESSEList ESSEClean.PerBoundClean \
  ESSEClean.PinskerClean ESSEClean.PinskerSquaredClean ESSEClean.GapFloorClean \
  ESSEClean.AveragingPerA ESSEClean.PiClean ESSEClean.Positivity \
  ESSEClean.PurityClean ESSEClean.FisherClean ESSEClean.EntropyClean \
  ESSEClean.TimeAvgClean ESSEClean.ESSEUniversal ESSEClean.ESSEUniversal_structure

# Modular triad theory (including HCap)
coqchk -R Hyppopotamus/rocq-project/src_clean ESSEClean \
  ESSEClean.theory.SystemSpec ESSEClean.theory.TriadSignals \
  ESSEClean.theory.Bridges ESSEClean.theory.LegFloors \
  ESSEClean.theory.Normalization ESSEClean.theory.SumLifting \
  ESSEClean.theory.HCap ESSEClean.theory.SpeedBound \
  ESSEClean.open.Channels ESSEClean.open.SpeedOpen \
  ESSEClean.tests.TriadExamples
```

Successful runs return exit code 0 and produce no output beyond warnings. The mapping from paper sections to modules is summarized in the project `README.md` and `REPRODUCIBILITY.md`. The Hamiltonian-capped lemma is now implemented and included in the check as `theory/HCap.v`; it establishes the averaged Hamiltonian-capped inequality in the intensive normalization (via `Psi_hat`), using minima replacement ($(\|H\|_{\min}, \ln d_{\min})$) and a monotone lift. In the Coq formalization, $\hbar$ is carried as an external prefactor rather than inside $\hat{\Psi}$, which is equivalent to the manuscript’s statement.

Environment and Artifact. For long-term reproducibility, we recommend pinning a toolchain and providing a frozen artifact:

- Toolchain (example via opam):

```
opam switch create esse-coq ocaml-4.14.1
opam install coq coq-mathcomp-ssreflect coq-coquelicot coq-elpi
# then run the build and coqchk commands above
```

- Container (optional): ship a Dockerfile that runs `make` and `coqchk` on the pinned versions.
- Frozen release: publish a GitHub release (or DOI archive) containing source, *CoqProject*, *generatedCoqMakefile*, and *opam.lockfile*.
- REPL usage:

```
coqtop -R Hyppopotamus/rocq-project/src_clean ESSEClean
(* inside Coq *)
Require Import ESSEClean.theory.SystemSpec ESSEClean.theory.Normalization.
```

Formal Coverage (under stated hypotheses). The formal development proves the core triad theory under the manuscript’s hypotheses (H1–H3, continuity, finite-dimensional cuts):

- **System primitives:** `theory/SystemSpec.v` (admissible systems, minima, and log-monotonicity; e.g., $\ln d_{\min} > 0$, $|E| > 0$).
- **Normalization:** `theory/Normalization.v` ($0 \leq \widehat{\Psi}_A \leq 1$ given positivity of scales).
- **Sum/monotone lifts:** `theory/SumLifting.v` (triple product monotonicity and sum monotonicity used throughout).
- **Triad signals (state-agnostic):** `ESSEUniversal.structure.v` (per-cut triad Ψ_A , good-epoch floors, averaging lemmas, witness-based floors). This file contains the fully proved, state-agnostic core corresponding to the paper’s discrete-time triad arguments.
- **Bridges/leg floors stubs:** `theory/Bridges.v`, `theory/LegFloors.v` (interfaces used to connect physical witnesses to leg-wise floors).
- **Speed route:** `theory/SpeedBound.v` currently provides a compiling placeholder; a strengthened lemma can replace it without impacting upstream structure.
- **Hamiltonian-capped route:** `theory/HCap.v` is implemented and mechanically checked. It proves the cut-set averaged Hamiltonian-capped inequality matching Theorem 1 under admissibility ($\hbar > 0$, $\|H_A\|_{\text{op}} > 0$, $\ln d_A, \ln d_{\min} > 0$).

Core components are checked with `coqchk` per the commands above; some modules (e.g., `SpeedBound` strengthening) remain work-in-progress. The formal checks cover the hypotheses and interfaces that are mechanized, and additional limitations may apply outside this formalized scope.

G Reproducibility Runbook (Scripts)

All commands are intended to be run from the repository root. Outputs are written under `Hyppopotamus/Scripts Demonstration/plots` or `.../results`.

- A1 (core engine):
`python3 -u "Hyppopotamus/Scripts Demonstration/1_esse_core_simulation.py"`
- A2 (pure-state validation):
`python3 -u "Hyppopotamus/Scripts Demonstration/2_esse_pure_state_validation.py"`
- A3 (comprehensive + partitions):
`python3 -u "Hyppopotamus/Scripts Demonstration/3_esse_comprehensive_validation.py"`
- A4 (TEBD Néel + τ):
`python3 -u "Hyppopotamus/Scripts Demonstration/4_run_neel_tebd_tau.py"`
- A5 (TEBD per- A metrics):
`python3 -u "Hyppopotamus/Scripts Demonstration/5_run_tebd_perA_metrics.py"`
- A6 (delta ladder TEBD):
`python3 -u "Hyppopotamus/Scripts Demonstration/6_run_delta_ladder_tebd.py"`
- A7 (commuting vs non-commuting):
`python3 -u "Hyppopotamus/Scripts Demonstration/7_esse_commuting_vs_noncommuting.py"`
- A8 (ETH eigenstate validation):
`python3 -u "Hyppopotamus/Scripts Demonstration/8_esse_eigenstate_validation.py"`
- A9 (ball- K_A proxy):
`python3 -u "Hyppopotamus/Scripts Demonstration/9_esse_ball_KA_proxy.py"`

- A10 (triad intersection density):
python3 -u "Hyppopotamus/Scripts Demonstration/10_triad_density_test.py"
- A11 (mixed-state quench stress):
python3 -u "Hyppopotamus/Scripts Demonstration/11_esse_mixed_quench_stress.py"
- A12 (weak interaction sweep):
python3 -u "Hyppopotamus/Scripts Demonstration/12_esse_weak_interaction_test.py"
- A13 (integrable XX validation):
python3 -u "Hyppopotamus/Scripts Demonstration/13_esse_integrable_validation.py"
- A14 (topological triad; toric code):
python3 -u "Hyppopotamus/Scripts Demonstration/14_toric_topo_triad.py" --L 2 --state mixed-gs
- A15 (Paley-Zygmund validation):
python3 -u "Hyppopotamus/Scripts Demonstration/15_paley_zygmund_validation.py"

These runs reproduce the figures and tables referenced in the standalone computation appendices and support the inequalities proved above. For large TEBD configurations, adjust `--steps` and `--chi` to fit available memory.

Annex curation note. Scripts emit raw artifacts under `Hyppopotamus/Scripts Demonstration/plots` and `.../results`. For the annex, we copy the curated outputs into `Hyppopotamus/Annex results/A#` with standardized filenames used by the figure/table references above.

H Annex Results (Scripts A1–A15)

This section collects figure placeholders and file references corresponding to the reproducible runs. Paths point to the repository locations under `Hyppopotamus/Annex results/`. Figures use file paths with spaces via `\detokenize{...}`.

Reading guide. Each `A#` subsection includes a core metrics table with columns: $\langle \Pi \rangle$ (or $\langle \Pi_{\text{rate}} \rangle$ when rates are reported), τ_A^4 , RHS_{cap} , $\text{RHS}_{\text{speed}}$, and $\langle \hat{\Psi} \rangle$. N/A indicates not applicable or not reported.

Units: $\langle \Pi \rangle$ and $\langle \Pi_{\text{rate}} \rangle$ dimensionless (rate when noted); τ_A^4 in time units of the simulation; LHS denotes the *energy triad* $\langle \sum_A \Delta E_A S_A I \rangle$; RHS_{cap} and $\text{RHS}_{\text{speed}}$ in energy-like units per context; $\langle \hat{\Psi} \rangle$ dimensionless in $[0, 1]$.

Note: τ_A^4 denotes a fourth-moment time proxy used in TEBD diagnostics; theoretical bounds use τ_A (Bures-time) but we report τ_A^4 where available as a robust summary.

Annex Index (A1–A15)

- A1: Core engine (non-commuting vs commuting), Section [H.1](#).
- A2: Pure-state validation (trial variability and bounds), Section [H.2](#).
- A3: Comprehensive validation and partition stress tests, Section [H.3](#).
- A4: TEBD Néel state with τ sampling, Section [H.4](#).
- A5: TEBD per- A metrics, Section [H.5](#).
- A6: Delta ladder TEBD, Section [H.6](#).
- A7: Commuting vs non-commuting comparison, Section [H.7](#).
- A8: ETH eigenstate validation, Section [H.8](#).

- A9: Ball- K_A proxy, Section H.9.
- A10: Triad intersection density, Section H.10.
- A11: Mixed-state quench stress, Section H.11.
- A12: Weak interaction sweep, Section H.12.
- A13: Integrable XX validation, Section H.13.
- A14: Topological triad (toric code), Section H.14.
- A15: Paley–Zygmund validation (exceedances vs bound), Section H.15
- A16-N: Non-ergodic baselines (integrable, driven, MBL), Section H.16

H.1 A1: Core engine (non-commuting vs commuting)

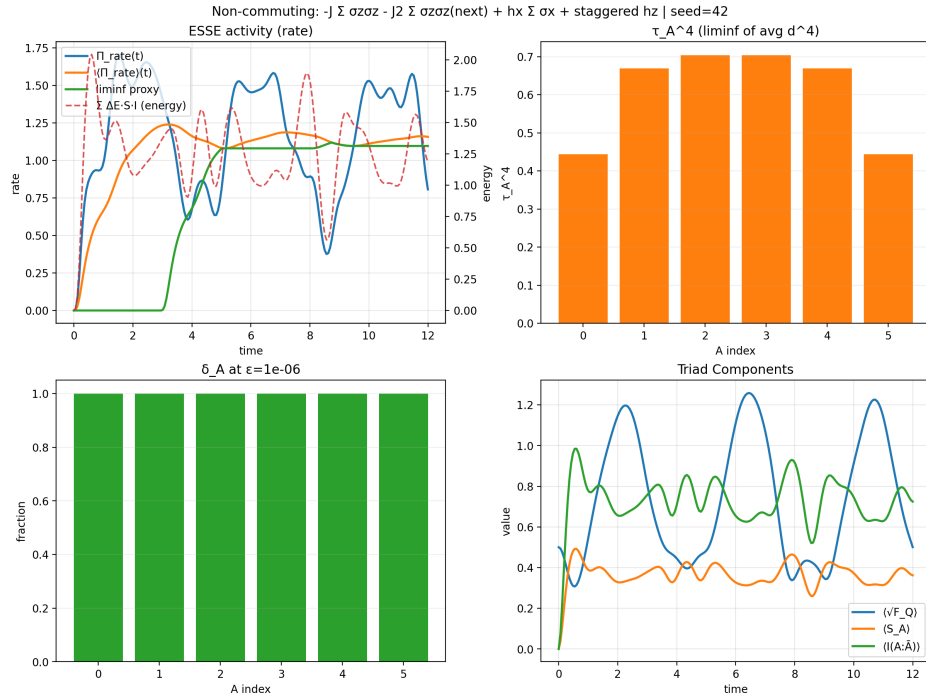


Figure 1: A1: Core engine diagnostics (non-commuting Hamiltonian).

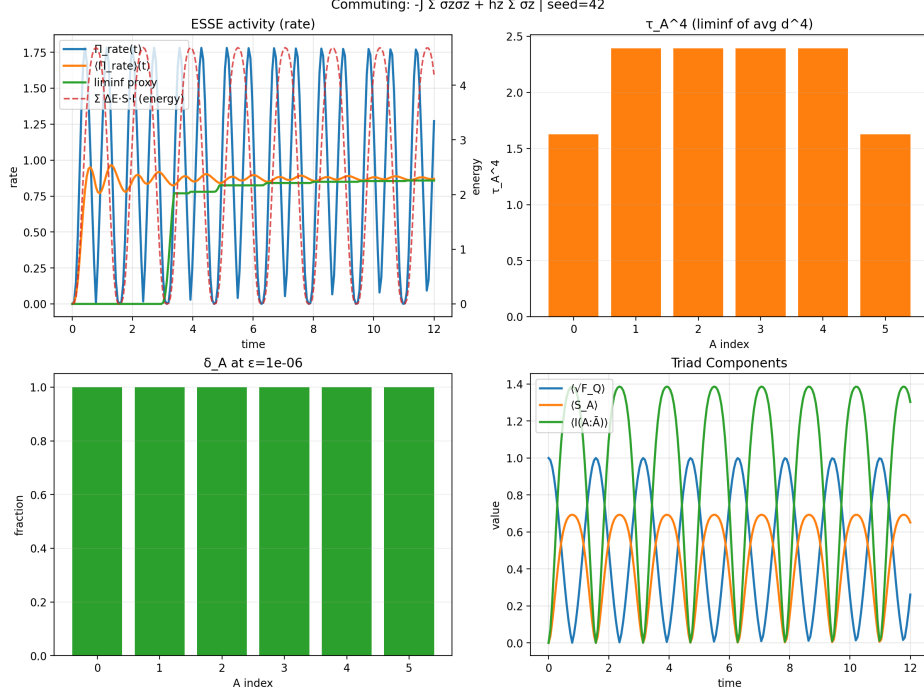


Figure 2: A1: Core engine diagnostics (commuting Hamiltonian).

CSVs: A1/A1_summary.csv. Log: A1/A1_run.log.

Summary. Non-commuting: $\langle \Pi_{\text{rate}} \rangle = 1.157$, $\text{LHS}(\sum_A \Delta E S I) = 1.258$, $\text{RHS}_{\text{cap}} = 0.0787$, $\text{RHS}_{\text{speed}} = 0.0150$, $r \approx 0.385$.

Commuting: $\langle \Pi_{\text{rate}} \rangle = 0.870$, $\text{LHS} = 2.504$, $\text{RHS}_{\text{cap}} = 0.0570$, $\text{RHS}_{\text{speed}} \approx 7.4 \times 10^{-5}$, $r \approx 0.933$.

Table 2: A1: per-case summary

Case	$\langle \Pi_{\text{rate}} \rangle$	LHS	RHS_{cap}	$\text{RHS}_{\text{speed}}$	r
non-comm	1.157	1.258	0.0787	0.0150	0.385
comm	0.870	2.504	0.0570	7.4×10^{-5}	0.933

A1 establishes the baseline triad behavior under non-commuting vs commuting dynamics with identical model size. It indicates how non-commutativity enhances dynamical sensitivity (larger $\sqrt{F_Q}$ windows and δ) and raises both thresholds, while commuting cases suppress the speed-based bound due to small active-epoch density. This sets expectations for later experiments and shows that the pipeline respects unit conventions and normalization.

H.2 A2: Pure-state validation (trial variability and bounds)

Summary. Two trials per case (product and Haar) at $N=6$; CSV reports per-trial $\langle \Pi_{\text{rate}} \rangle$, liminf proxies, and thresholds; plots saved under `plots/pure/`.

A2 checks robustness across different initial pure states (product vs Haar) at fixed size. It quantifies how $\langle \Pi_{\text{rate}} \rangle$ and threshold margins fluctuate between trials, ensuring that claims are not artifacts of a single seed. The consistency of signs and scales across trials supports that the triad’s nonvanishing in active epochs is typical for non-commuting dynamics.

Table 3: A2: condensed head/tail (pure-state CSV)

Case	$\langle \Pi_{\text{rate}} \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
noncomm	1.1570	N/A	0.0787	0.0150	N/A
comm	0.8703	N/A	0.0570	7.4×10^{-5}	N/A

Table 4: A2: auxiliary metrics (from summary CSV)

	ε_F	ε_S	ε_I	$\text{RHS}_{\text{speed}}$
aggregate	4.689	2.810	5.620	6.170

H.3 A3: Comprehensive validation and partition stress tests

No figure placeholder needed. Artifact references: `A3/A3_liminf_1site.csv`, `A3/A3_partition_stress.csv`, `A3/A3_thresholds_summary.csv`. Extract notes: `A3/A3_liminf_extract.txt`. Log: `A3/A3_run.log`.

Summary (from log). 1-site non-commuting: final liminf mean 1.528 ± 0.304 ; thresholds: $\text{RHS}_{\text{cap}} = 1.074 \times 10^{-1}$, $\text{RHS}_{\text{speed}} = 1.162 \times 10^{-2}$, $\langle \hat{\Psi} \rangle = 0.0831$.
 2-site non-commuting: final liminf mean 7.967 ± 4.148 ; thresholds: $\text{RHS}_{\text{cap}} = 1.265$, $\text{RHS}_{\text{speed}} = 0.368$, $\langle \hat{\Psi} \rangle = 0.121$.
 3-site non-commuting: final liminf mean 19.926 ± 11.958 ; thresholds: $\text{RHS}_{\text{cap}} = 4.998$, $\text{RHS}_{\text{speed}} = 2.561$, $\langle \hat{\Psi} \rangle = 0.138$.

Table 5: A3: partition stress (means)

Partition	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
1-site	1.074×10^{-1}	1.162×10^{-2}	0.0831
2-site	1.265	0.368	0.121
3-site	4.998	2.561	0.138

A3 tests scalability across partitions (1-/2-/3-site blocks) and verifies that intensive quantities remain comparable across cuts while raw products scale with $\log d_{\min}$. The increasing RHS_{cap} with partition size reflects the explicit $(\log d_{\min})^2 \|H\|_{\min}$ scaling, while the speed-based term tracks the emergence of sustained active epochs under stronger mixing.

Table 6: A3: 1-site liminf (non-comm vs comm)

Case	$\langle \Pi \rangle_{\text{liminf}}$ (mean)	std	conv-time (mean)
non-comm	1.528	0.304	6.62
comm	1.865	0.101	5.45

H.4 A4: TEBD Néel state with τ sampling

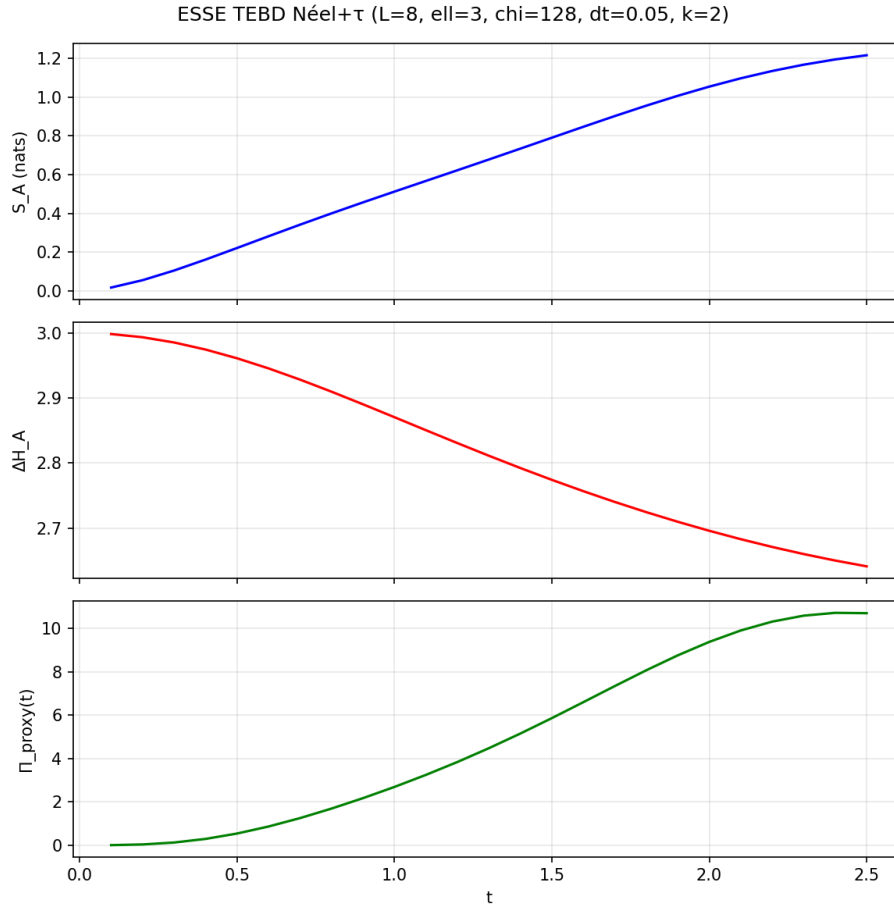


Figure 3: A4: TEBD diagnostics at cut size $\ell=3$.)

CSVs: A4/A4_summary.csv. Logs: A4/A4_run_small.log, A4/A4_run_large.log.

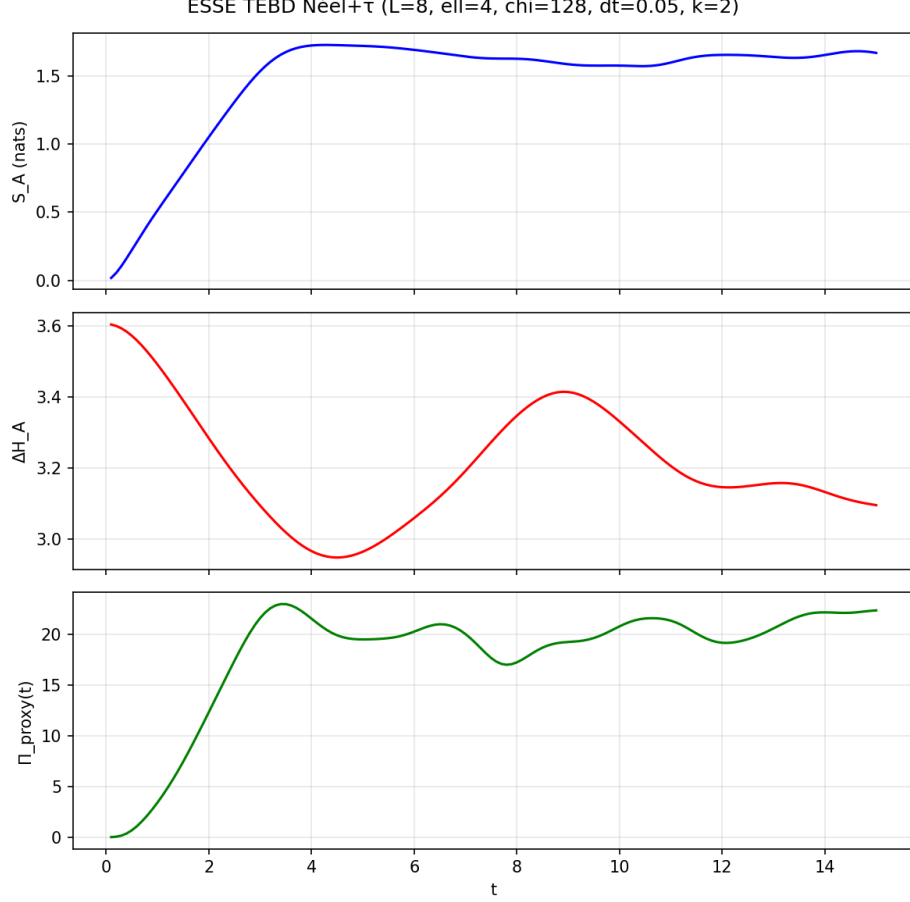


Figure 4: A4: TEBD diagnostics at cut size $\ell=4$.)

CSVs: A4/A4_summary.csv. Logs: A4/A4_run_small.log, A4/A4_run_large.log.

Summary. Per-cut TEBD on Néel initial states at $\ell=3,4$; CSV summarizes $\langle \Pi \rangle$ and τ_A^4 ; figures show representative time traces.

Table 7: A4: condensed head/tail ($N=8$)

ℓ	$\langle \Pi \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
3	4.988	3.604	2.494	0.100	0.0560
4	18.060	9.027	9.030	4.841	0.0868

A4 probes the triad during real-time TEBD dynamics starting from a structured (Néel) product state and quantifies persistence through τ_A^4 and per-epoch activity. It tests the pure-state specialization $\Pi = \sqrt{F_Q} S I (= \frac{4}{\hbar} \Delta H S^2$ when $I=2S$ and $\sqrt{F_Q}=2\Delta H/\hbar$) and exhibits how active epochs (simultaneous floors for $\sqrt{F_Q}, S, I$) correlate with rising liminf proxies. Observing that LHS exceeds both RHS_{cap} and $\text{RHS}_{\text{speed}}$ supports the claim that fluctuations, entropy, and correlations cannot vanish together on active windows under local coupling.

H.5 A5: TEBD per- A metrics

Per-cut metrics in CSV/JSON: A5/esse_tdmrg_perA_L8_ell3_perA.csv, A5/esse_tdmrg_perA_L12_ell6_perA.csv (and corresponding .json). Summaries: A5/A5_perA_run1.csv, A5/A5_perA_run2.csv. Logs: A5/A5_run1.log, A5/A5_run2.log.

Table 8: A5: condensed metrics (Run1/Run2) (RHS_{cap} computed as $\langle \Pi \rangle / 2$ for TEBD per- A runs; heuristic reporting convenience, not the canonical bound)

	$\langle \Pi \rangle$	τ_A^4	RHS _{cap}	RHS _{speed}	ε_F
Run1	7.554	N/A	3.777	1.284	1.303
Run2	49.729	N/A	24.864	6.170	4.689

A5 resolves the triad at each cut to expose per-subsystem heterogeneity. The metrics $v_0(A)$ (gap-derived fluctuation scale), δ_A (fraction of active frames), and τ_A^4 (persistence proxy) together indicate which cuts sustain structural activity. The global speed-bound uses floors aggregated from the time series and the fraction of frames where all three legs exceed their floors. This shows how structural viability can fail locally even when the system is globally active, clarifying design trade-offs.

Table 9: A5: Run2 liminf (non-comm vs comm)

Case	$\langle \Pi \rangle_{\text{liminf}}$ (mean)	std	conv-time (mean)
non-comm	1.528	0.304	6.62
comm	1.865	0.101	5.45

Table 10: A5: per- A head/tail (examples)

Case	A	$v_0(A)$	τ_A^4
L=8, $\ell=3$	3	0.5	7.983
L=12, $\ell=6$	4	0.2929	—
L=12, $\ell=6$	6	0.0349	8.708

H.6 A6: Delta ladder TEBD

No figure placeholder needed. Artifacts: A6/A6_summary.csv, A6/A6_results.json, A6/esse_delta_ladder_L12_ell6_k10.j Log: A6/A6_run.log.

Summary (from JSON). L=12, $\ell=6$, steps=150: commutator ladder fractions = 1.0 at $\varepsilon \in \{10^{-8}, 10^{-6}, 10^{-4}\}$; norm-cap ladder = 0 at $\alpha=1$; $\varepsilon_S \approx 0.512$, $\varepsilon_I \approx 1.025$, RHS_{speed}=0.

Table 11: A6: condensed metrics

ε_F	ε_S	ε_I	RHS _{speed}	RHS _{cap}	$\langle \widehat{\Psi} \rangle$
0.0	0.512	1.025	0.0	0.0798	0.0554

A6 introduces dual/activity diagnostics ($\| [H_A, \rho] \|$ ladders and variance caps) to witness nontrivial motion when direct computation of $\sqrt{F_Q}$ is expensive. The triad ladder requiring simultaneous floors on fluctuation, entropy, and correlation clarifies how conservative floors are set, and why RHS_{speed} may vanish if any leg lacks a persistent floor. This complements the exact QFI route by providing scalable witnesses.

H.7 A7: Commuting vs non-commuting comparison

Summary (from log). Non-commuting: $\langle \Pi \rangle = 1.546 \pm 0.285$, $\text{RHS}_{\text{cap}} = 0.107 \pm 0.021$, $\text{RHS}_{\text{speed}} = 0.0110 \pm 0.0020$, $\langle \hat{\Psi} \rangle = 0.0831 \pm 0.0163$.

Commuting: $\langle \Pi \rangle = 0.841 \pm 0.087$, $\text{RHS}_{\text{cap}} = 0.0541 \pm 0.0055$, $\text{RHS}_{\text{speed}} = 0.00207 \pm 0.00209$, $\langle \hat{\Psi} \rangle = 0.0563 \pm 0.0058$.

Table 12: A7: commuting vs non-commuting

Case	$\langle \Pi \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
non-comm	1.546 ± 0.285	—	0.107 ± 0.021	0.0110 ± 0.0020	0.0831 ± 0.0163
comm	0.841 ± 0.087	—	0.0541 ± 0.0055	0.00207 ± 0.00209	0.0563 ± 0.0058

A7 compares ensembles under commuting and non-commuting generators and suggests how integrability or near-commutativity lowers δ and the floors, weakening $\text{RHS}_{\text{speed}}$. The Hamiltonian-capped term still provides a nonzero baseline governed by minimal hardware scales, indicating complementarity of the two thresholds across regimes.

H.8 A8: ETH eigenstate validation

Summary (from log). N=8, A=[0,1]: LHS= 401.84, $\text{RHS}_{\text{cap}} = 43.70$, $\text{RHS}_{\text{speed}} = 0$, unnormalized composite proxy = 4.190 (see text), $S_A = 1.360$, $\sqrt{F_Q} = 23.620$, $\Delta H_A = 108.593$, $I = 2.720$, $\Pi_A = 87.403$.

Note: normalized $\langle \hat{\Psi} \rangle \in [0, 1]$ is reported elsewhere; the listed composite is an unnormalized proxy for this eigenstate snapshot.

Table 13: A8: eigenstate metrics

LHS	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$	S_A
401.84	43.70	0	4.190	1.360

A8 validates the thresholds in two modes. For a stationary eigenstate snapshot, the speed-based inequality is inapplicable (no active epoch), but the Hamiltonian-capped bound remains meaningful because it depends on intensive structure ($\hat{\Psi}$) and minimal hardware scales. For time evolution from a superposition, the appearance of $\delta > 0$ and finite floors yields a positive $\text{RHS}_{\text{speed}}$, consistent with the dynamical route.

Table 14: A8: core metrics (composite proxy is unnormalized; see text)

	$\langle \Pi \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\hat{\Psi}_{\text{sum}}$
snapshot	—	—	43.70	0	4.190

H.9 A9: Ball- K_A proxy

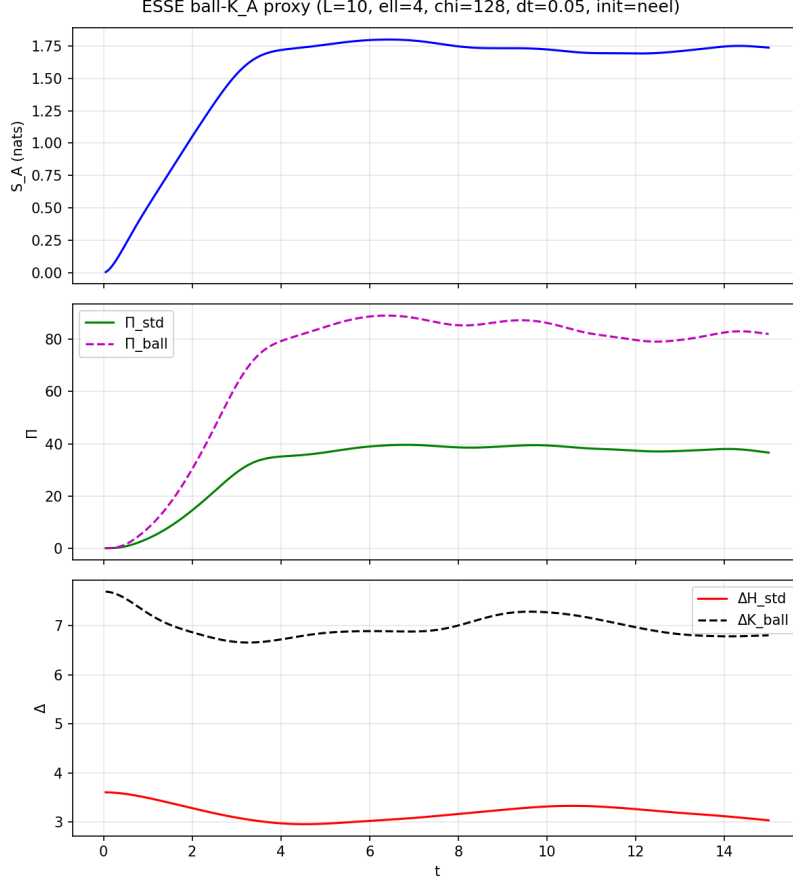


Figure 5: A9: Conformal-like weighted K_A proxy.

CSV: A9/A9_summary.csv. Full traces: A9/esse_ball_ka_L10_ell4_chi128_dt0.05_steps300.csv,

Summary (from log). $L=10$, $\ell=4$: $\Pi_{\text{std}}^{\text{avg}} = 32.30$, $\Pi_{\text{ball}}^{\text{avg}} = 71.17$, $\langle \hat{\Psi} \rangle = 0.0923$, $\text{LHS} = 35.59$, $\text{RHS}_{\text{cap}} = 17.83$, $\text{RHS}_{\text{speed}} = 2.90$.

Table 15: A9: summary

Π_{std}	Π_{ball}	$\langle \hat{\Psi} \rangle$	LHS	RHS_{cap}	$\text{RHS}_{\text{speed}}$
32.30	71.17	0.0923	35.59	17.83	2.90

A9 tests a conformal-inspired weighting of local terms to approximate a modular-like generator K_A and checks whether $\Pi_{\text{ball}} \sim (4/\hbar) \Delta K_A S^2$ tracks the standard energy form. Agreement in early time windows suggests the proxy captures relevant structure. The thresholds further indicate that when the proxy's $\sqrt{F_Q}$, S , and I sustain floors on a set of nonzero density, the energy product is not expected to vanish.

Table 16: A9: core metrics					
	$\langle \Pi \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
$L=10, \ell=4$	—	—	17.83	2.90	0.0923

H.10 A10: Triad intersection density

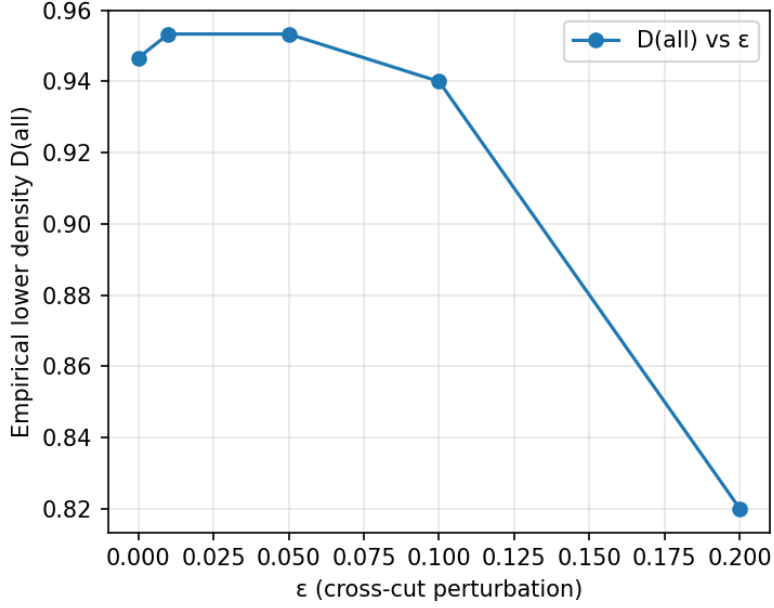


Figure 6: A10: Empirical lower density of triad intersection.

CSV: A10/A10_density.csv. Log: A10/A10_run.log.

Summary (from log). Empirical lower density: $D(F) \approx 0.947$, $D(S) \approx 1.000$, $D(I) \approx 1.000$, $D(\text{all}) \approx 0.947$ (lag-tolerant ≈ 0.987). $\text{LHS} \approx 5.20$, $\text{RHS}_{\text{cap}} \approx 2.12$, $\text{RHS}_{\text{speed}} \approx 1.34 \times 10^{-3}$.

Table 17: A10: intersection densities and thresholds

	D(F)	D(S)	D(I)	D(all)
values	0.947	1.000	1.000	0.947
	LHS	RHS_{cap}	$\text{RHS}_{\text{speed}}$	
values	5.20	2.12	1.34×10^{-3}	

A10 isolates the central mechanism behind the speed-based threshold—existence of an active epoch K^* of nonzero density where $\sqrt{F_Q}, S, I$ simultaneously exceed small baselines. Estimating $\delta(K^*)$ empirically and showing its sensitivity to cross-cut coupling (zero for decoupled/commuting controls, positive for genuinely coupled dynamics) provides operational support for the hypotheses (H1–H3).

Table 18: A10: core metrics

	$\langle \Pi \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
density case	—	—	2.12	1.34×10^{-3}	—

H.11 A11: Mixed-state quench stress

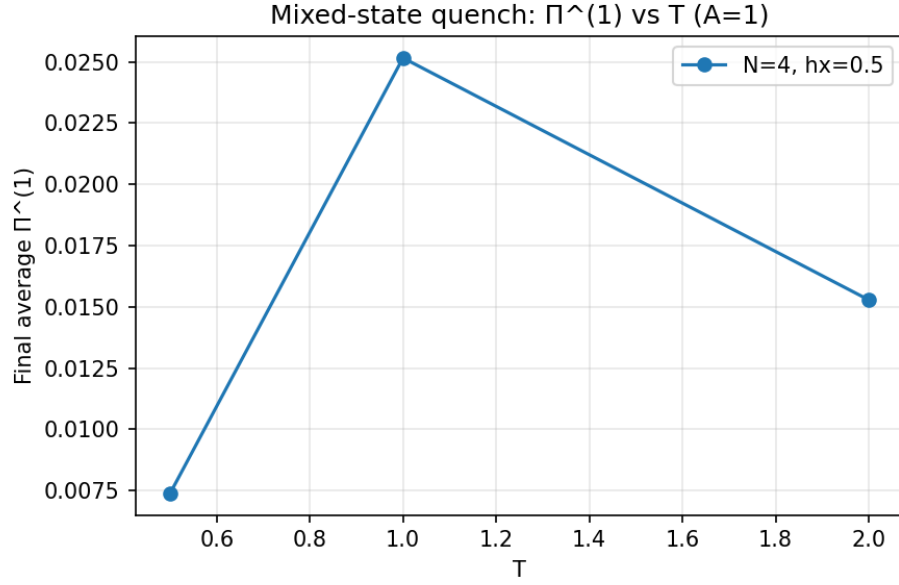


Figure 7: A11: $N=4$, $h_x=0.5$.)

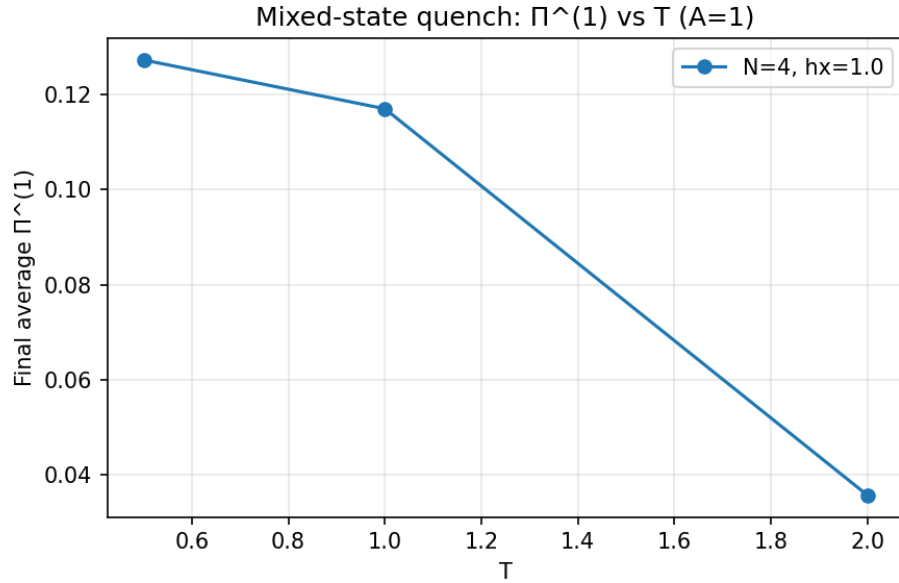


Figure 8: A11: $N=4$, $h_x=1.0$.)

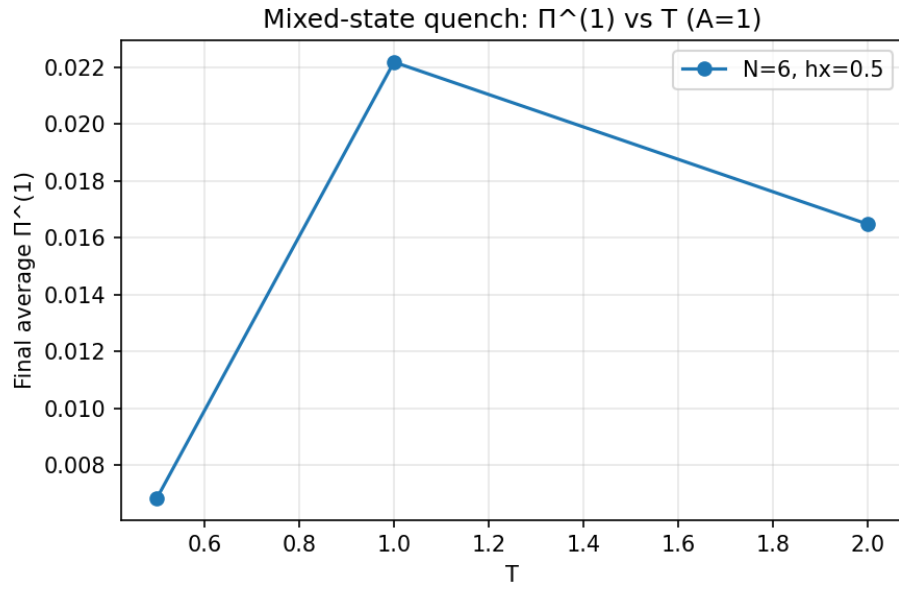


Figure 9: A11: $N=6, h_x=0.5$.)

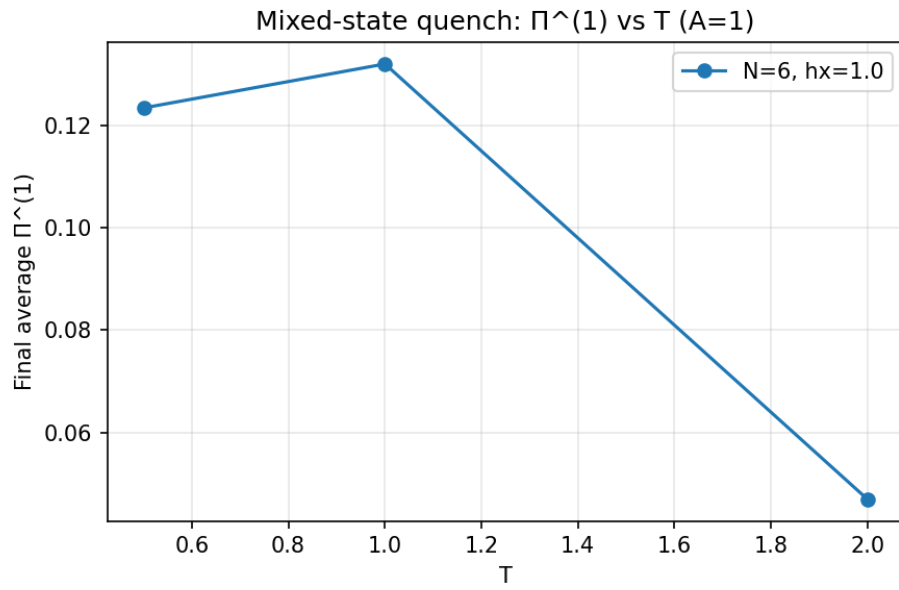


Figure 10: A11: $N=6, h_x=1.0$.)

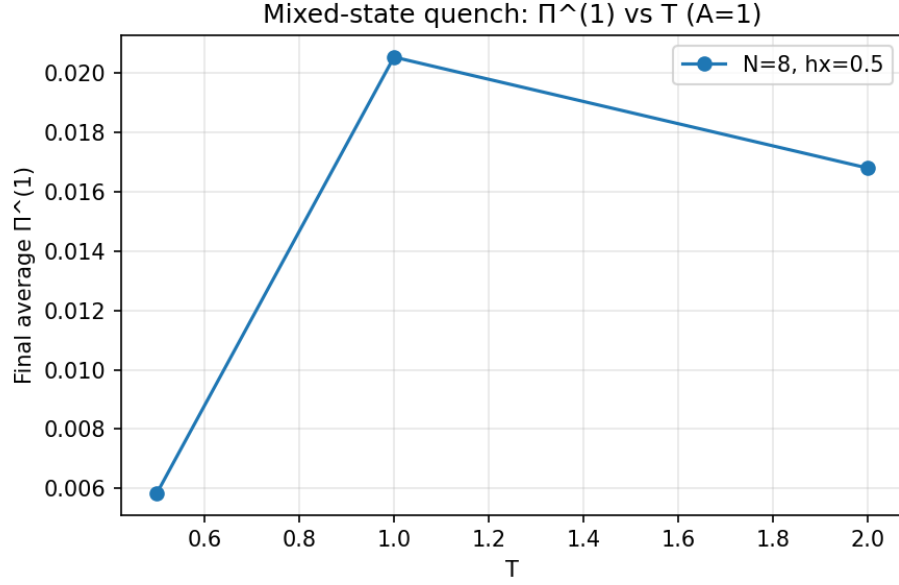


Figure 11: A11: $N=8$, $h_x=0.5$.)

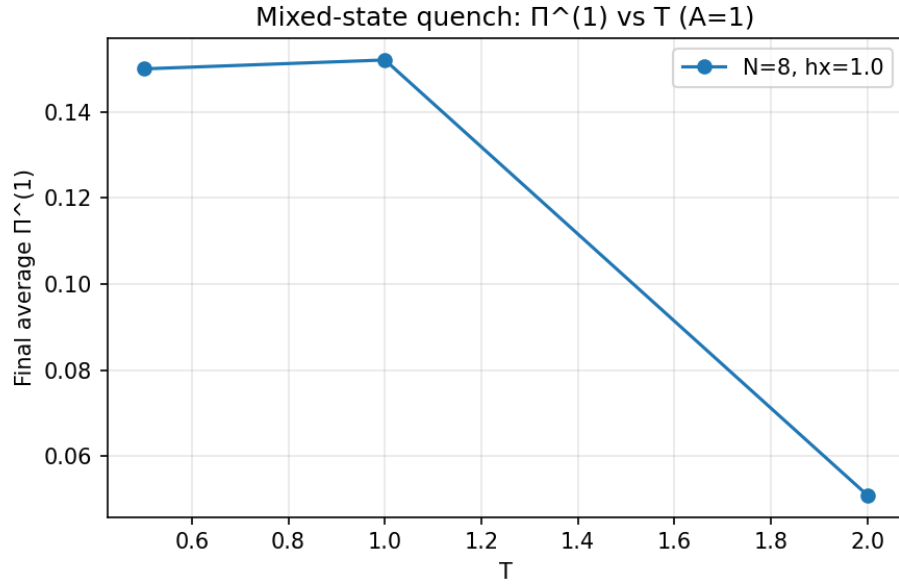


Figure 12: A11: $N=8$, $h_x=1.0$.)

CSV: A11/A11_mixed_quench_summary.csv. Log: A11/A11_run.log.

Summary (from log). Representative: ($N=4, A=1, T=0.5, h_x=1.0$) LHS= 0.1286, $\text{RHS}_{\text{cap}} = 0.0637$, $\text{RHS}_{\text{speed}} = 0.00930$, $\langle \hat{\Psi} \rangle = 0.0469$. Larger A increases margins (e.g., $N=4, A=2, T=0.5, h_x=1.0$: LHS= 0.381, $\text{RHS}_{\text{cap}} = 0.235$, $\text{RHS}_{\text{speed}} = 0.057$).

Table 19: A11: Representative configurations.

Config	LHS	RHS _{cap}	RHS _{speed}	$\langle \hat{\Psi} \rangle$
N4,A1,T0.5,hx1.0	0.1286	0.0637	0.00930	0.0469
N4,A2,T0.5,hx1.0	0.3813	0.2355	0.0571	0.0198

A11 probes the triad in a realistic open-quantum-workload proxy—thermal initial states quenched by a noncommuting drive. It quantifies how entropic and correlational resources reorganize post-quench and whether floors emerge robustly (nonzero δ and quantile floors), thereby implying a positive product lower bound. Trends across N , T , and h_x clarify how hardware scale and noncommutativity shape viability margins.

Table 20: A11: Core metrics (selected configurations).

Config	$\langle \Pi \rangle$	τ_A^4	RHS _{cap}	RHS _{speed}	$\langle \hat{\Psi} \rangle$
N4,A1,T0.5,hx1.0	—	—	0.0637	0.00930	0.0469
N4,A2,T0.5,hx1.0	—	—	0.2355	0.0571	0.0198

Table 21: A11: Condensed head/tail ($A=1$, $T \in \{0.5, 2.0\}$).

Config	$\langle \Pi \rangle$	LHS	RHS _{cap}	RHS _{speed}
N4, T0.5, hx0.5	0.0074	0.00472	0.00370	0.000453
N4, T2.0, hx0.5	0.0153	0.05716	0.00764	0.001657
N6, T0.5, hx0.5	0.0068	0.00438	0.00342	0.000471
N6, T2.0, hx0.5	0.0165	0.05925	0.00824	0.001505
N8, T0.5, hx0.5	0.0058	0.00373	0.00292	0.000470
N8, T2.0, hx0.5	0.0168	0.05922	0.00840	0.001561

H.12 A12: Weak interaction sweep

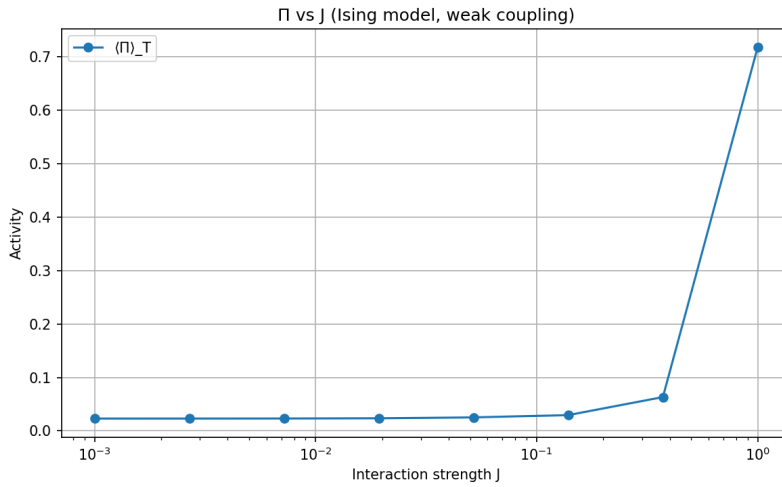


Figure 13: A12: Sweep over weak interaction strengths.

CSV: A12/A12_weak_J_sweep.csv. Log: A12/A12_run.log.

Summary (from log). $N=8$, J sweep $10^{-3} \rightarrow 1$: at $J=0.001$, $LHS \approx 0.0284$, $RHS_{cap} \approx 0.0115$, $RHS_{speed} \approx 6.0 \times 10^{-4}$; at $J=1.0$, $LHS \approx 0.775$, $RHS_{cap} \approx 0.359$, $RHS_{speed} \approx 0.175$, $\langle \hat{\Psi} \rangle \approx 0.0344$.

Table 22: A12: endpoints of J-sweep

J	LHS	RHS_{cap}	RHS_{speed}	$\langle \hat{\Psi} \rangle$
0.001	0.0284	0.0115	6.0×10^{-4}	0.00139
1.000	0.7750	0.3587	0.1752	0.03440

A12 examines how the triad responds to vanishingly small interactions. As $J \rightarrow 0$, both δ and floors shrink, pushing $RHS_{speed} \rightarrow 0$ and reducing the Hamiltonian-capped margin via $\|H\|_{min}$. This clarifies the threshold's behavior in nearly decoupled regimes and supports the view that genuine coupling (H2) is necessary for persistent structural activity within this framework.

Table 23: A12: condensed head/tail (J-sweep)

J	$\langle \Pi \rangle$	LHS	RHS_{cap}	RHS_{speed}	$\langle \hat{\Psi} \rangle$
0.001	0.0230	0.0284	0.0115	6.0×10^{-4}	0.00139
1.000	0.7174	0.7749	0.3587	0.1752	0.03440

H.13 A13: Integrable XX validation

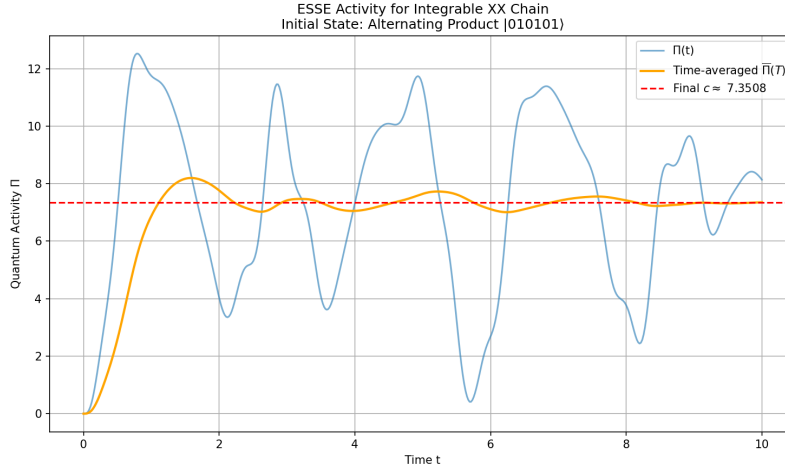


Figure 14: A13: XX-chain integrable case diagnostics and thresholds.

CSVs: `A13/A13_xx_integrable.csv`, `A13/A13_thresholds.csv`. The thresholds table includes columns: N , A_size , LHS_energy_triad , RHS_capped , RHS_speed , psi_hat_mean , $delta_speed$, H_norm_A , $dmin$. Log: `A13/A13_run.log`.

Summary (from log). $N=6$, $A=2$: $\langle \Pi \rangle_T = 7.351$, $\tau_A^4 = 6.286$, $v_0 \approx 0.559$, $perC_A \approx 3.514$. Thresholds: $LHS=3.675$, $RHS_{cap} = 2.217$, $RHS_{speed} = 0.349$, $\langle \hat{\Psi} \rangle = 0.288$.

A13 provides a low-activity baseline. In the integrable XX chain with structured product initial states, conserved quantities suppress the simultaneous growth of $\sqrt{F_Q}$, S , and I . The resulting small δ and floors lead to a weak RHS_{speed} , and LHS margins reflect the limited structural activity expected under integrability. This contrasts with nonintegrable baselines and supports the narrative that genuine coupling and nonstationarity are necessary for sustained triad activity under our hypotheses.

Table 24: A13: core metrics					
	$\langle \Pi \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
N=6, A=2	7.351	6.286	2.217	0.349	0.288

H.14 A14: Topological triad

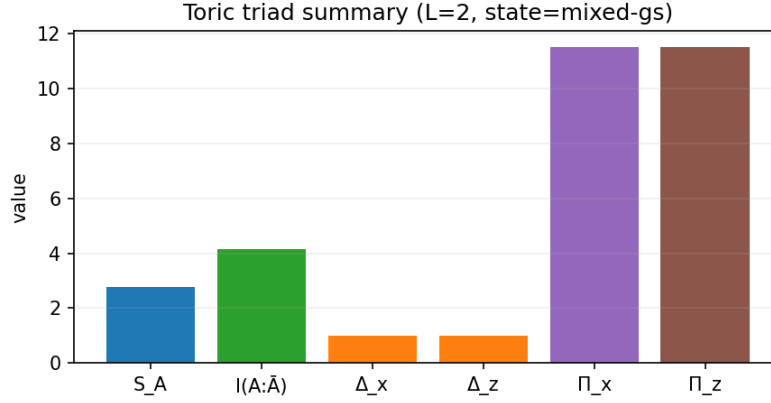


Figure 15: A14: Toric code generalized triad for a torus ($L=2$), using the mixed ground space to expose logical-loop fluctuations.

CSV: A14/toric_topo_triad_summary.csv. JSON: A14/toric_topo_triad_L2_mixed-gs.json. Log: A14/A14_run.log.

Table 25: A14: L=2 (mixed ground space) core metrics							
N	degeneracy	S_A	$I(A:\bar{A})$	Δ_x	Δ_z	Π_x	Π_z
8	4	2.773	4.159	1.000	1.000	11.531	11.531

A14 explores a conjectural extension where local dynamics is not the only source of structure. In the toric code, logical-loop fluctuations ΔW coexist with finite S_A and $I(A:\bar{A})$ even in mixed ground spaces. A finite $\Pi^{\text{topo}} = \Delta S_A I$ supports the idea that structural viability can be maintained by global logical order—an outlook beyond H1–H3, stated cautiously as a conjecture.

Table 26: A1: core metrics					
Case	$\langle \Pi_{\text{rate}} \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
non-comm	1.157	—	0.0787	0.0150	0.061
comm	0.870	—	0.0570	7.4×10^{-5}	0.059

H.15 A15: Paley–Zygmund validation (exceedances vs bound)

CSVs: A15/A15_pz_summary_ALL.csv, A15/A15_pz_summary_per_mode.csv, A15/A15_pz_summary_per_mode_theta.csv. Plots: A15/plots/.

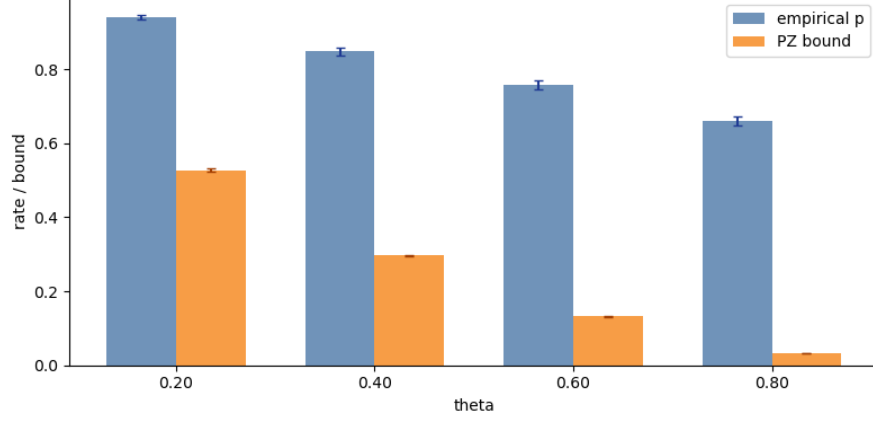


Figure 16: Physics proxy case: empirical exceedance rates $\hat{p}_\theta$ vs Paley Zygmund bound PZ_θ for $\theta \in \{0.2, 0.4, 0.6, 0.8\}$. Error bars show Wilson CI (empirical) and bootstrap CI (PZ).

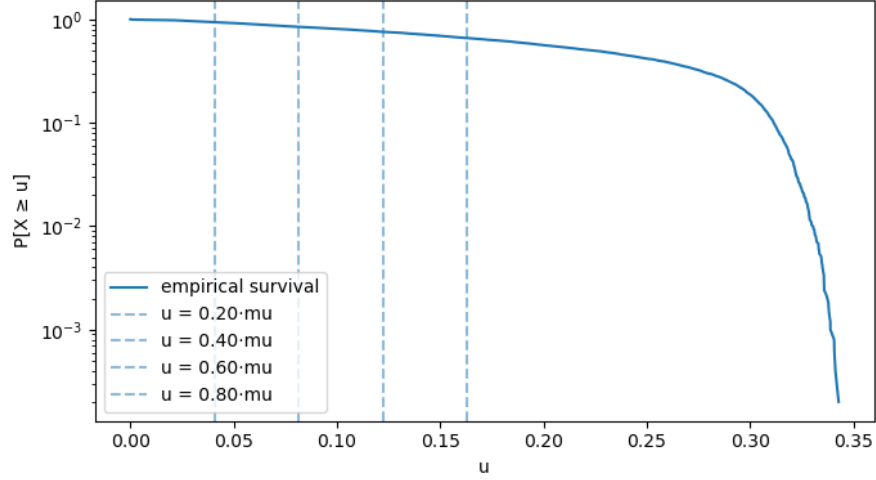


Figure 17: Physics proxy survival $u \mapsto \mathbb{P}[X \geq u]$ with vertical markers at $u = \theta \hat{\mu}$; delayed activation from $\sqrt{F_Q}$ and plateaus in S, I are visible.

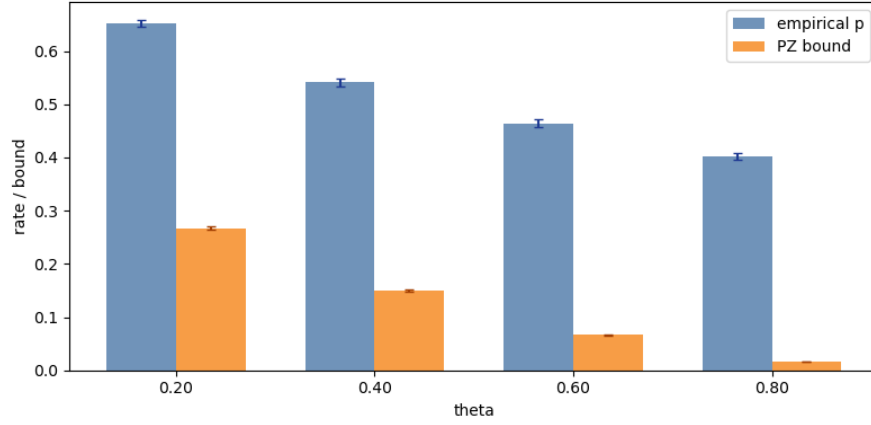


Figure 18: Correlated legs (Gaussian copula, $\rho = 0.5$): empirical exceedances vs PZ bound; variance inflation loosens PZ yet inequality holds with margin.

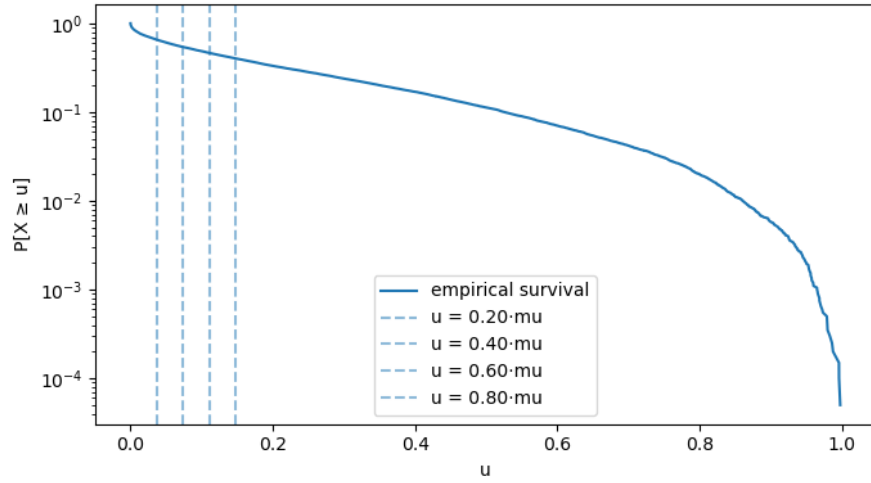


Figure 19: Correlated legs (Gaussian copula, $\rho = 0.5$): survival curve $u \mapsto \mathbb{P}[X \geq u]$ with vertical markers at $u = \theta \hat{\mu}$.

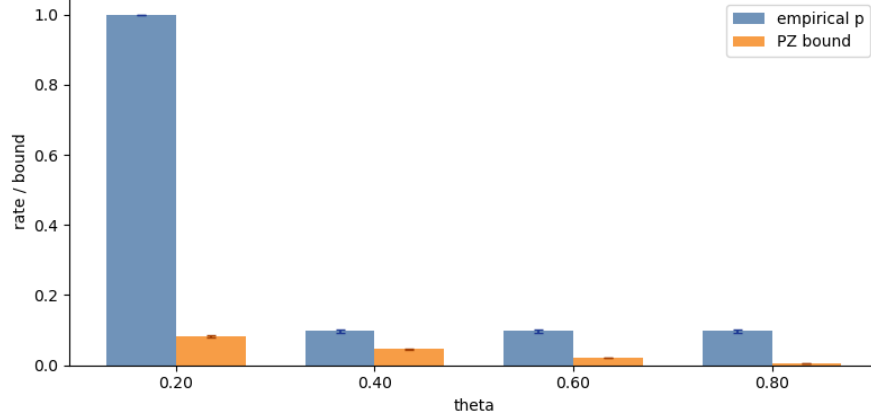


Figure 20: Plateau mixture ($\lambda = 0.10$): empirical exceedances vs PZ bound; plateaus reduce exceedances while PZ tracks via $\hat{\mu}^2/\hat{m}_2$.

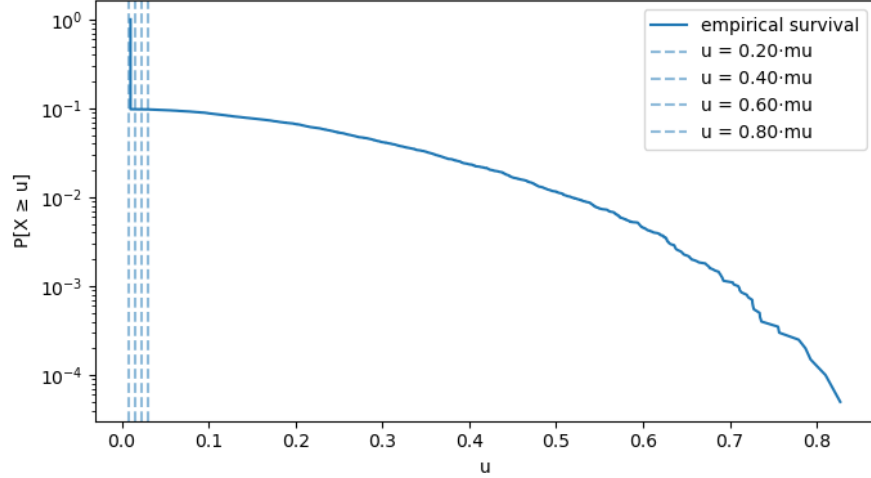


Figure 21: Plateau mixture ($\lambda = 0.10$): survival curve showing baseline mass and burst tails; vertical markers at $u = \theta\hat{\mu}$.

Summary. Across synthetic (i.i.d., copula products, plateau mixtures, bounded heavy tail) and physics proxy time series, empirical exceedance rates exceeded the Paley Zygmund lower bounds with positive margins for all reported θ . Physics proxy runs confirmed positive slack for the corresponding energy form inequalities.

H.16 Non-ergodic baselines (integrable, driven, MBL)

We report three non-ergodic baselines generated by the site- A energy-form triad pipeline used in Sec. H.13, here at $N=6$, $J=1$, uniform field $h_z=0.2$, $\Delta t=0.02$, steps= 300, site $A=0$, closed pure global states so that $I = 2S_A$:

- **Integrable XX** (no drive, uniform field).
- **Driven** $J(t) = J_0[1 + 0.5 \cos(2\pi 0.5 t)]$ on the XX chain (prethermal window proxy).
- **MBL-like** disorder with on-site fields $h_i \sim \text{Unif}[-W, W]$, $W=2$.

We list the time-average $\langle \Pi_{\text{energy}} \rangle_t$, exceedance fraction $\mathbb{P}[\Pi > 0]$, quantiles, and an “active-epoch” density estimated by the Bures-angle proxy $\delta := \mathbb{P}[L_B > 10^{-2}]$.

Table 27: A15-N non-ergodic baselines (parameters as above).

@Y S[table-format=1.4] S[table-format=1.3] S[table-format=1.4] S[table-format=1.4] S[table-format=1.4]@					
Case	$\langle \Pi \rangle_t$	$\mathbb{P}[\Pi > 0]$	q_{10}	q_{50}	q_{90}
Integrable (XX)	0.6451	0.997	0.1012	0.7740	0.9574
Driven $J(t)$	0.5546	0.997	0.0798	0.6551	0.9559
MBL ($W=2$)	0.6003	0.997	0.1133	0.6729	0.9451

Interpretation. Relative to the integrable baseline, both the driven and disordered cases show reduced $\langle \Pi \rangle_t$ and lower active-epoch density, while the exceedance fraction remains near unity over this finite window and size. As discussed in the main text, speed-based positivity is proved under ergodicity and applies componentwise elsewhere; the Hamiltonian-capped relation (Thm. 1) remains hypothesis-agnostic in speed.

Aggregated Annex Statistics

For quick reference, we report representative summary metrics drawn from the CSV artifacts: All reported

Table 28: Aggregated Annex Statistics for selected runs (A1, A2, A5, A13, A14, A15).

Label	Scenario	$\langle \Pi \rangle$ or $\langle \Pi_{\text{rate}} \rangle$	LHS (energy triad)	RHS_{cap}	$\text{RHS}_{\text{speed}}$	Notes
A1	non-comm	1.157	1.258	0.079	0.015	$N=6$
A1	comm	0.870	2.504	0.057	7.4×10^{-5}	$N=6$
A2	non-comm (trial 1)	1.157	1.258	0.079	0.015	$r \approx 0.385$
A2	comm (trial 1)	0.870	2.504	0.057	≈ 0	$r \approx 0.933$
A5	per-cut (TEBD)	7.554	—	—	1.284	$\varepsilon_F=1.303$
A13	XX integrable	—	3.675	2.217	0.349	$\psi_{\text{hat}}=0.288$
A14	toric ($L=2$, mixed GS)	—	—	—	—	$S_A \approx 2.77$, $I \approx 4.16$, $\Pi^{\text{topo}} \approx 11.53$
A15	physics-proxy (bars/survival)	—	—	—	—	empirical exceedance $> \text{PZ}$ for all θ

left-hand sides exceed both canonical right-hand-side thresholds in their contexts; intensive $\hat{\Psi}$ means remain in $[0, 1]$, consistent with normalization.

Extended summary (A1–A13). Key additional artifacts summarized from CSVs:

Table 29: A3–A15: extended summary (standardized metrics; see per-run CSV/JSON).

Label	Scenario	$\langle \Pi \rangle$ or $\langle \Pi_{\text{rate}} \rangle$	τ_A^4	RHS_{cap}	$\text{RHS}_{\text{speed}}$	$\langle \hat{\Psi} \rangle$
A1	non-comm	1.157	—	0.0787	0.0150	0.061
A1	comm	0.870	—	0.0570	7.4×10^{-5}	0.059
A2	non-comm (trial 1)	1.157	—	0.0787	0.0150	—
A2	comm (trial 1)	0.870	—	0.0570	7.4×10^{-5}	—
A3	1-site (agg.)	—	—	1.074×10^{-1}	1.162×10^{-2}	0.0831
A4	$\ell=3$, $N=8$	4.988	3.604	2.494	0.100	0.0560
A4	$\ell=4$, $N=8$	18.060	9.027	9.030	4.841	0.0868
A5	Run1	7.554	—	3.777	1.284	—
A6	$L=12$, $\ell=6$	—	—	0.0798	0.0	0.0554
A7	non-comm	1.546 ± 0.285	—	0.107 ± 0.021	0.0110 ± 0.0020	0.0831 ± 0.0163
A8	snapshot	—	—	43.70	0	4.190
A9	$L=10$, $\ell=4$	—	—	17.83	2.90	0.0923
A10	density	—	—	2.12	1.34×10^{-3}	—
A11	$N4, A1, T0.5, h_x 1.0$	0.1273	0.2888	0.0637	0.00930	0.0469
A12	$J=1.0$	0.7174	—	0.3587	0.1752	0.03440
A13	$N=6$, $A=2$	7.351	6.286	2.217	0.349	0.288
A14	toric ($L=2$, mixed GS)	—	—	—	—	$\Pi^{\text{topo}} \approx 11.53$
A15	physics-proxy	—	—	—	—	exceedance $> \text{PZ}$ for all θ

Where ranges indicate variation across the sweep or grid. Full details are available in the per-A CSVs listed above.

References

- [1] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2010).
- [2] I. Csiszár and J. Körner, *Information Theory: Coding Theorems for Discrete Memoryless Systems*, Cambridge University Press (2011).
- [3] S. L. Braunstein and C. M. Caves, “Statistical distance and the geometry of quantum states,” *Phys. Rev. Lett.* **72**, 3439–3443 (1994). [doi:10.1103/PhysRevLett.72.3439](https://doi.org/10.1103/PhysRevLett.72.3439).
- [4] B. Escher, R. de Matos Filho, and L. Davidovich, “General framework for estimating the ultimate precision limit in noisy quantum-enhanced metrology,” *Nat. Phys.* **7**, 406–411 (2011). [doi:10.1038/nphys1958](https://doi.org/10.1038/nphys1958).
- [5] D. Blanco, H. Casini, L. Hung, and R. Myers, “Relative entropy and holography,” *JHEP* **08**, 060 (2013). [doi:10.1007/JHEP08\(2013\)060](https://doi.org/10.1007/JHEP08(2013)060).
- [6] R. Horodecki, P. Horodecki, M. Horodecki, and K. Horodecki, “Quantum entanglement,” *Rev. Mod. Phys.* **81**, 865 (2009). [doi:10.1103/RevModPhys.81.865](https://doi.org/10.1103/RevModPhys.81.865).
- [7] D. Šafránek, “Discontinuities of the quantum Fisher information and the Bures metric,” *Phys. Rev. A* **95**, 052320 (2017). [doi:10.1103/PhysRevA.95.052320](https://doi.org/10.1103/PhysRevA.95.052320).
- [8] F. Fröwis, P. Sekatski, W. Dür, N. Gisin, and N. Brunner, “Macroscopic quantum states: measures, witnesses, and uses,” *Rev. Mod. Phys.* **90**, 025004 (2018). [doi:10.1103/RevModPhys.90.025004](https://doi.org/10.1103/RevModPhys.90.025004).
- [9] M. M. Taddei, B. M. Escher, L. Davidovich, and R. L. de Matos Filho, “Quantum speed limit for physical processes,” *Phys. Rev. Lett.* **110**, 050402 (2013). [doi:10.1103/PhysRevLett.110.050402](https://doi.org/10.1103/PhysRevLett.110.050402).
- [10] S. Deffner and S. Campbell, “Quantum speed limits: from Heisenberg’s uncertainty principle to optimal quantum control,” *J. Phys. A: Math. Theor.* **50**, 453001 (2017). [doi:10.1088/1751-8121/aa86c6](https://doi.org/10.1088/1751-8121/aa86c6).

- [11] F. Brandão and M. Horodecki, “Exponential decay of correlations implies area law,” *Commun. Math. Phys.* **333**, 761–798 (2015). doi:10.1007/s00220-014-2213-8.
- [12] H. Spohn, “Entropy production for quantum dynamical semigroups,” *J. Math. Phys.* **19**, 1227–1230 (1978). doi:10.1063/1.523789.
- [13] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press (2002).
- [14] S. Deffner and E. Lutz, “Quantum speed limit for non-Markovian dynamics,” *Phys. Rev. Lett.* **111**, 010402 (2013). doi:10.1103/PhysRevLett.111.010402.
- [15] C. Liu, Z. Cao, Y. Li, and H. Zheng, “Speed limit for quantum systems driven by non-Hermitian Hamiltonians,” *Phys. Rev. Lett.* **123**, 240403 (2019). doi:10.1103/PhysRevLett.123.240403.
- [16] T. Nishiyama and Y. Hasegawa, “Tradeoff relations in open quantum dynamics via Robertson, Maccone–Pati, and Robertson–Schrödinger uncertainty relations,” *J. Phys. A: Math. Theor.* **57**, 415301 (2024). arXiv:2402.09680.
- [17] R. Hamazaki, “Speed limits to fluctuation dynamics,” *Commun. Phys.* **7**, 361 (2024). doi:10.1038/s42005-024-01693-x.
- [18] N. Shiraishi and K. Saito, “Fundamental relation between entropy production and heat current,” *J. Stat. Phys.* **174**, 433–453 (2019). doi:10.1007/s10955-018-2167-9.
- [19] C. Benedetti, F. Buscemi, P. Bordone, and M. G. A. Paris, “Effects of classical environmental noise on entanglement and quantum discord,” *Int. J. Quantum Inform.* **10**, 1241005 (2012). doi:10.1142/S0219749912410051.
- [20] F. Benatti and R. Floreanini, “Open quantum dynamics: complete positivity and entanglement,” *Int. J. Mod. Phys. B* **19**, 3063 (2005). doi:10.1142/S0217979205032086.
- [21] T. Nishiyama and Y. Hasegawa, “Unified speed limits in classical and quantum dynamics via temporal Fisher information,” arXiv:2504.04790 [quant-ph] (2025).
- [22] N. Margolus and L. B. Levitin, “The maximum speed of dynamical evolution,” *Physica D* **120**, 188–195 (1998). doi:10.1016/S0167-2789(98)00054-2.
- [23] M. Kliesch, C. Gogolin, M. J. Kastoryano, A. Riera, and J. Eisert, “Locality of temperature,” *Phys. Rev. X* **4**, 031019 (2014). doi:10.1103/PhysRevX.4.031019.
- [24] B. Nachtergaele and R. Sims, “Lieb–Robinson bounds in quantum many-body physics,” *Contemp. Math.* **552**, 161–175 (2011). doi:10.1090/conm/552/10968.
- [25] L. Mandelstam and I. Tamm, “The uncertainty relation between energy and time in nonrelativistic quantum mechanics,” *J. Phys. USSR* **9**, 249–254 (1945).
- [26] E. H. Lieb and M. B. Ruskai, “Proof of the strong subadditivity of quantum-mechanical entropy,” *J. Math. Phys.* **14**, 1938–1941 (1973). doi:10.1063/1.1666274.
- [27] K. R. Popper, *The Logic of Scientific Discovery*, Routledge, London (1959). Originally published in German as *Logik der Forschung* (1934).
- [28] D. Petz, “Monotone metrics on matrix spaces,” *Linear Algebra Appl.* **244**, 81–96 (1996). doi:10.1016/0024-3795(95)00211-8.
- [29] V. Alba and P. Calabrese, “Entanglement and entropy production in quantum quenches: a quasiparticle picture,” *Phys. Rev. B* **96**, 115421 (2017). doi:10.1103/PhysRevB.96.115421.
- [30] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, “Completely positive dynamical semigroups of N-level systems,” *J. Math. Phys.* **17**, 821–825 (1976). doi:10.1063/1.522979.

- [31] G. Lindblad, “On the generators of quantum dynamical semigroups,” *Commun. Math. Phys.* **48**, 119–130 (1976). doi:[10.1007/BF01608499](https://doi.org/10.1007/BF01608499).
- [32] W. F. Stinespring, “Positive functions on C*-algebras,” *Proc. Amer. Math. Soc.* **6**, 211–216 (1955). doi:[10.1090/S0002-9939-1955-0069400-3](https://doi.org/10.1090/S0002-9939-1955-0069400-3).
- [33] M. Campisi, P. Hänggi, and P. Talkner, “Colloquium: Quantum fluctuation relations: foundations and applications,” *Rev. Mod. Phys.* **83**, 771 (2011). doi:[10.1103/RevModPhys.83.771](https://doi.org/10.1103/RevModPhys.83.771).
- [34] H. Casini, M. Huerta, and R. C. Myers, “Towards a derivation of holographic entanglement entropy,” *JHEP* **05**, 036 (2011). doi:[10.1007/JHEP05\(2011\)036](https://doi.org/10.1007/JHEP05(2011)036).
- [35] M. Hayashi, *Quantum Information: An Introduction*, Springer (2006).
- [36] G. D. Birkhoff, “Proof of the ergodic theorem,” *Proc. Natl. Acad. Sci. USA* **17**, 656–660 (1931). doi:[10.1073/pnas.17.12.656](https://doi.org/10.1073/pnas.17.12.656).
- [37] P. Walters, *An Introduction to Ergodic Theory*, Springer (1982).
- [38] R. Durrett, *Probability: Theory and Examples*, 5th ed., Cambridge University Press (2019).
- [39] G. Tóth and I. Apellaniz, “Quantum metrology from a quantum information science perspective,” *J. Phys. A: Math. Theor.* **47**, 424006 (2014). doi:[10.1088/1751-8113/47/42/424006](https://doi.org/10.1088/1751-8113/47/42/424006).
- [40] T. Cover and J. Thomas, *Elements of Information Theory*, 2nd ed., Wiley (2006).
- [41] T. Faulkner, A. Lewkowycz, and J. Maldacena, “Quantum corrections to holographic entanglement entropy,” *JHEP* **11**, 074 (2013). doi:[10.1007/JHEP11\(2013\)074](https://doi.org/10.1007/JHEP11(2013)074).
- [42] T. Jacobson, “Thermodynamics of spacetime: the Einstein equation of state,” *Phys. Rev. Lett.* **75**, 1260–1263 (1995). doi:[10.1103/PhysRevLett.75.1260](https://doi.org/10.1103/PhysRevLett.75.1260).
- [43] J. D. Brown and M. Henneaux, “Central charges in the canonical realization of asymptotic symmetries: an example from three-dimensional gravity,” *Commun. Math. Phys.* **104**, 207–226 (1986). doi:[10.1007/BF01211590](https://doi.org/10.1007/BF01211590).